

School of Computer Science and Engineering, UNSW



Week 01-02 : Introduction to Computer Networks

Computer Networks and Applications, COMP3331/9331

Never Stand Still

Nadeem Ahmed

Reading Guide: Chapter 1, Sections 1.1 - 1.4

Acknowledgement

- Majority of lecture slides are from the author's * lecture slide set
 - With enhancements + additional material

*Jim Kurose & Keith Ross
Computer Networking: A Top Down Approach

Introduction: Roadmap

- What is the Internet?
- What is a protocol?
- Network edge: hosts, access network, physical media
- Network core: packet/circuit switching, internet structure
- Performance: loss, delay, throughput
- Protocols layers, service models
- Security (self study, not on exam)
- History (self study, not on exam)

Hobbe's Internet Timeline - <http://www.zakon.org/robert/internet/timeline/>

Quiz 1-1

What is the Internet?

- A. One single homogenous network
- B. An interconnection of different computer networks
- C. An infrastructure that provides services to networked applications
- D. Something else



The Internet : a “nuts and bolts” view



Billions of connected computing **devices**:
Hosts = end systems
Running network **apps** at the Internet's edge



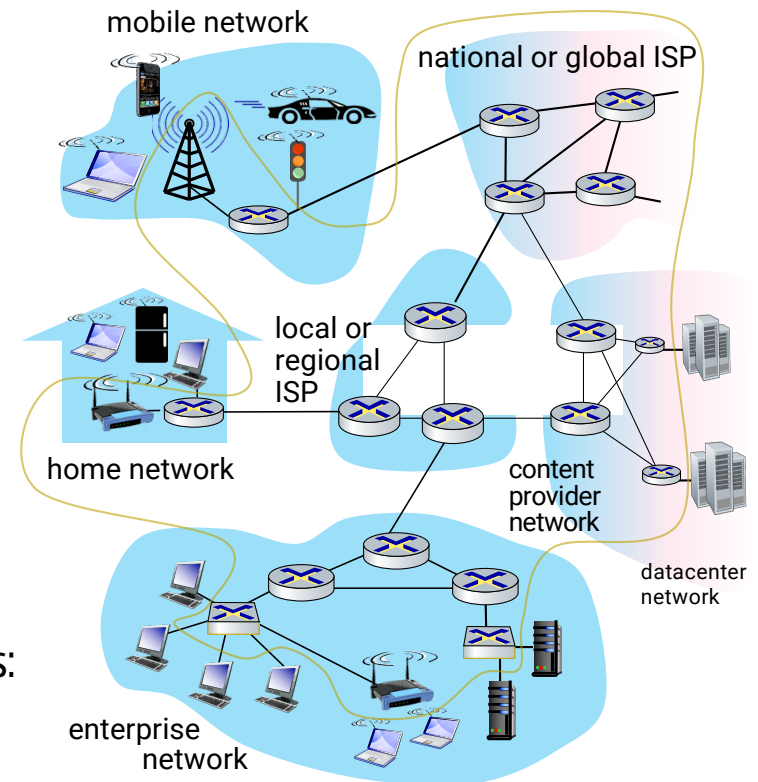
Packet switches: forward packets (chunks of data)
Routers, switches



Communication links
Fiber, copper, radio, satellite
Transmission rate: bandwidth



Networks
Collection of devices, routers, links:
managed by an organization



“Fun” Internet appliances



Security Camera



Picture frame



Web-enabled toaster +
weather forecaster



car



pacemaker



Tweet-a-watt:
monitor energy use



Fitbit



Amazon Echo



Internet
refrigerator



Networked TV Set top Boxes



sensorized,
bed
mattress

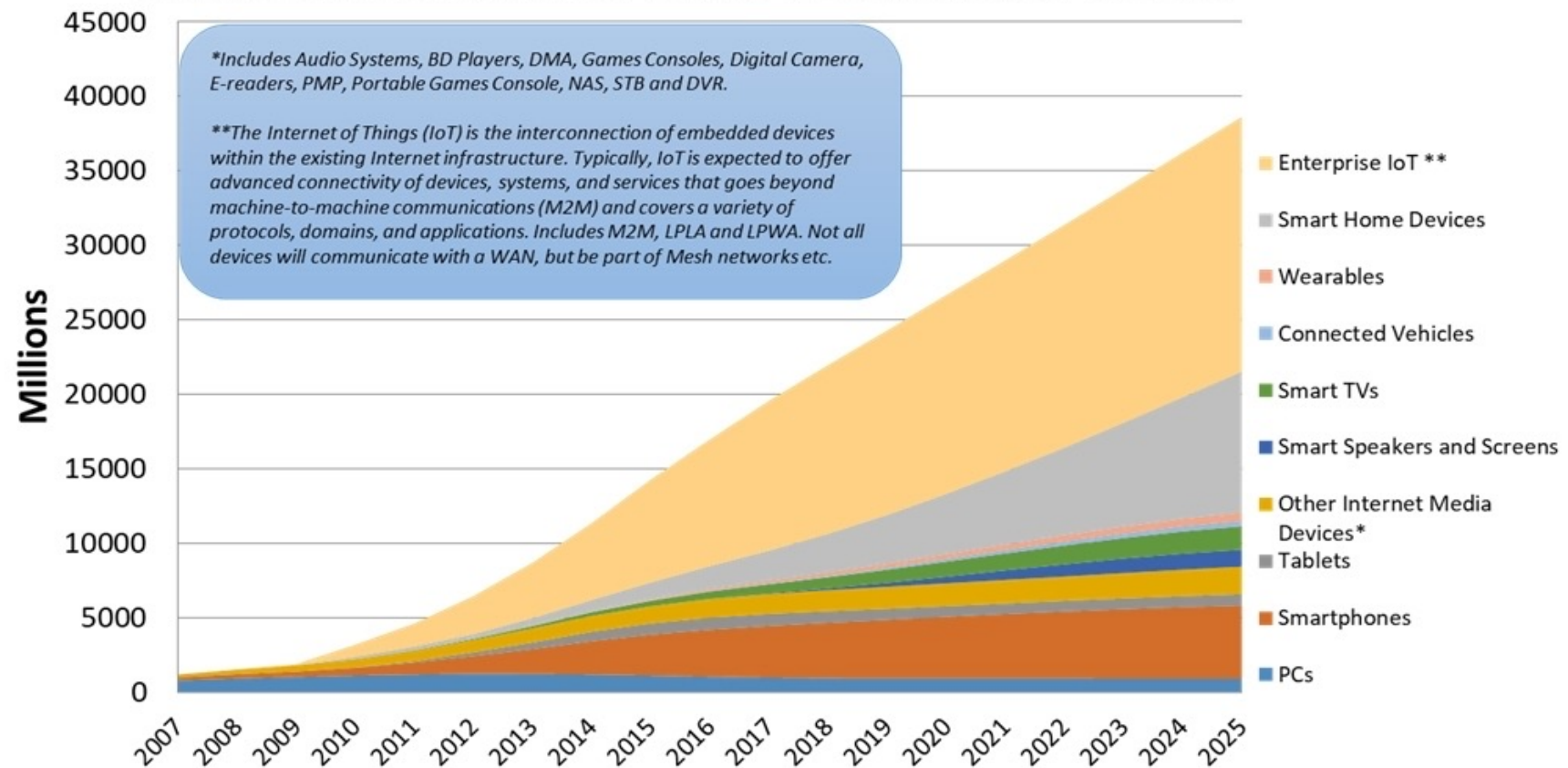


Smart Lightbulbs



AR devices

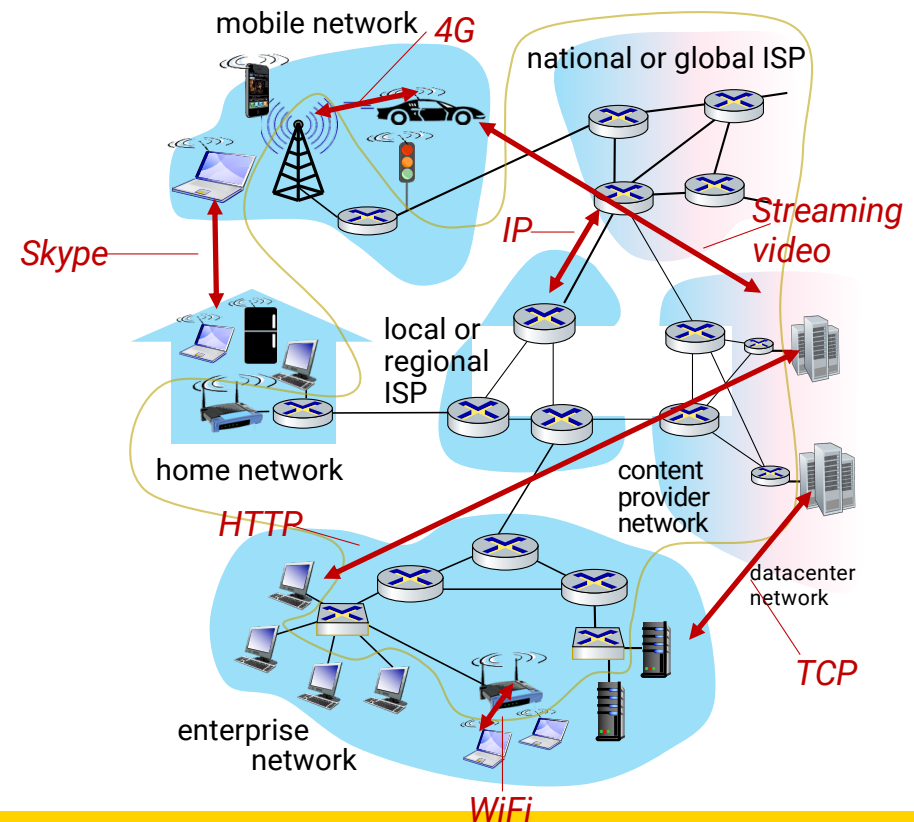
Global Connected and IoT Device Installed Base Forecast



Source – Strategy Analytics research services, May 2019: IoT Strategies, Connected Home Devices, Connected Computing Devices, Wireless Smartphone Strategies, Wearable Device Ecosystem, Smart Home Strategies

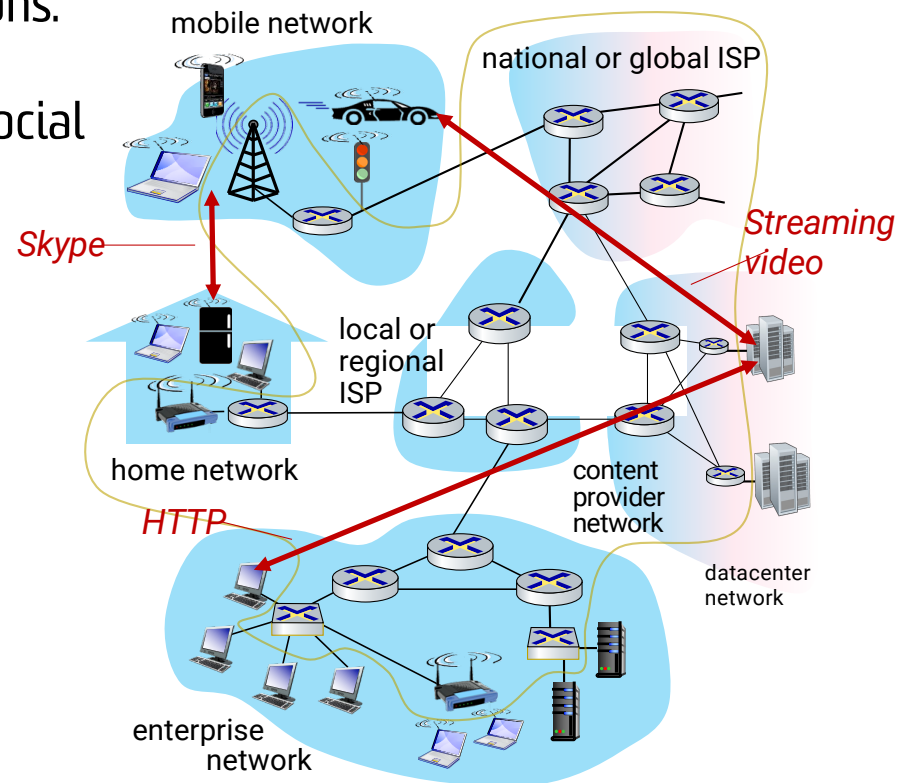
The Internet : a “nuts and bolts” view

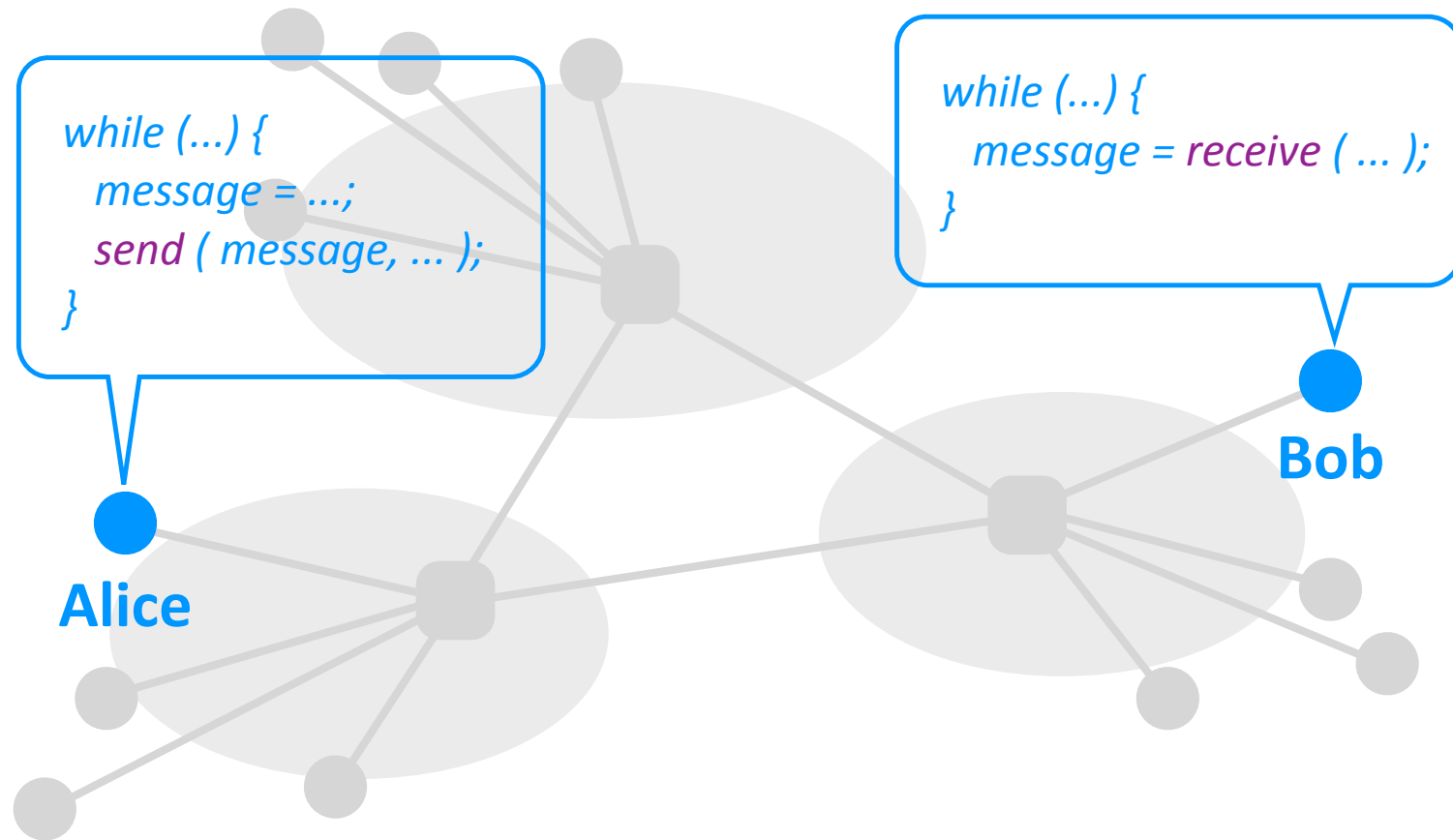
- **Internet: “network of networks”**
 - Interconnected ISPs
- **Protocols** are everywhere
 - Control sending, receiving of messages
 - e.g., HTTP (Web), streaming video, Skype, TCP, IP, WiFi, 4G, Ethernet
- **Internet standards**
 - RFC: Request for Comments
 - IETF: Internet Engineering Task Force

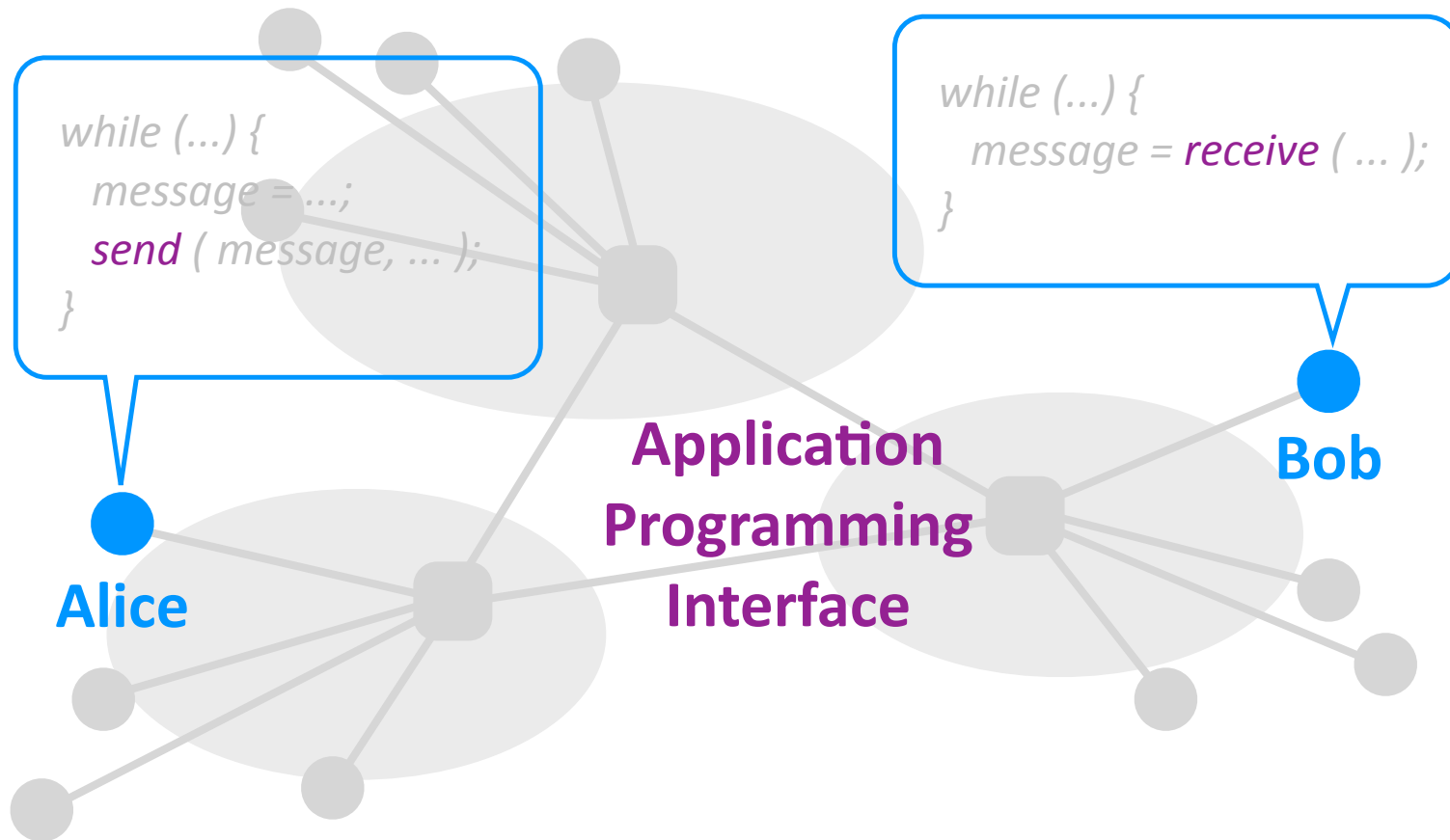


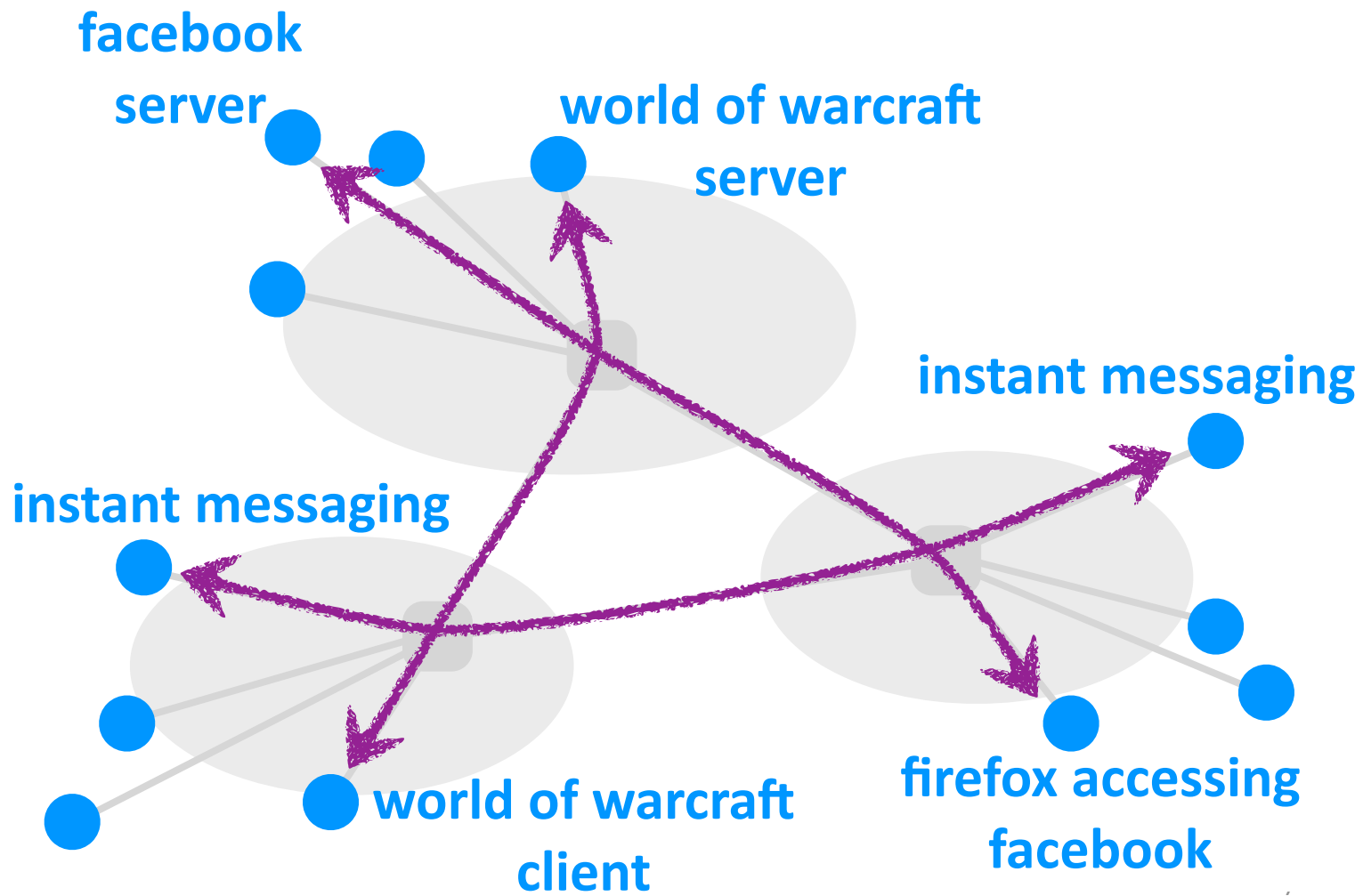
The Internet : a “service” view

- **Infrastructure** that provides services to applications:
 - Web, streaming video, multimedia teleconferencing, email, games, e-commerce, social media, inter-connected appliances, ...
- Provides **programming interface** to distributed applications:
 - “hooks” allowing sending/receiving apps to “connect” to, use Internet transport service
 - provides service options, analogous to postal service









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What is a Protocol?

Human protocols:

- “what’s the time?”
- “I have a question”
- introductions

... specific messages sent

... specific actions taken
when message
received, or other
events

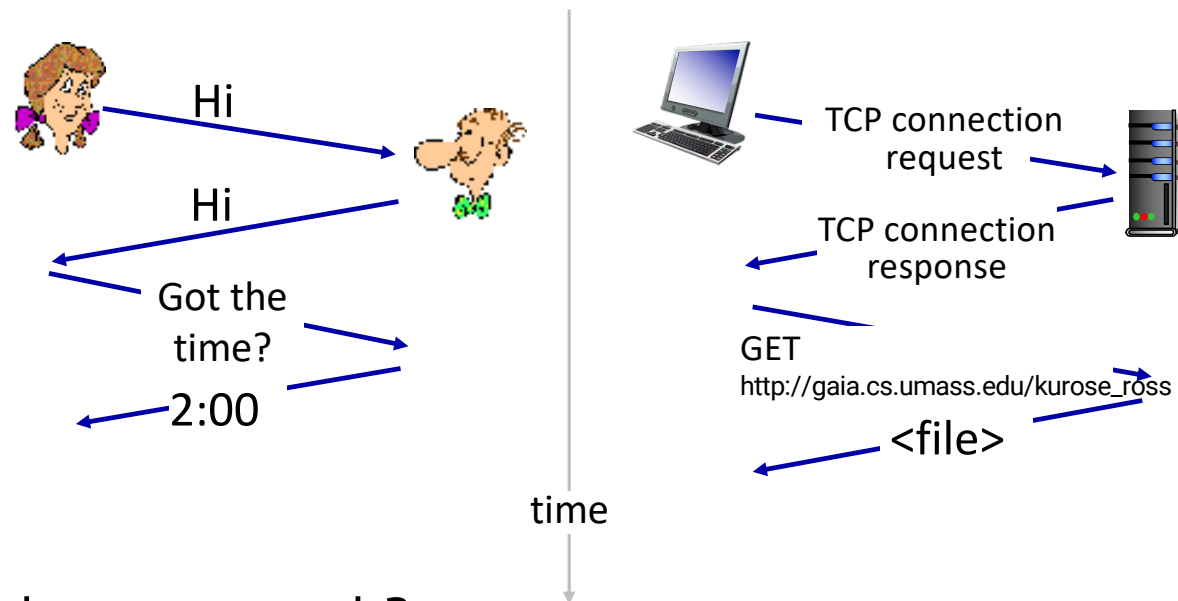
Network protocols:

- computers (devices) rather than humans
- all communication activity in Internet governed by protocols

*Protocols define the **format, order** of **messages sent and received** among network entities, and **actions taken** on msg transmission, receipt*

What is a Protocol?

A human protocol and a computer network protocol:



Q: Other human protocols?

Quiz 1-2

- How many Internet-connected devices do you have in your home (include your computers, phones, tablets, smart IoT devices)?



- A. Less than 10
- B. 10 to 20
- C. 21 to 50
- D. More than 50

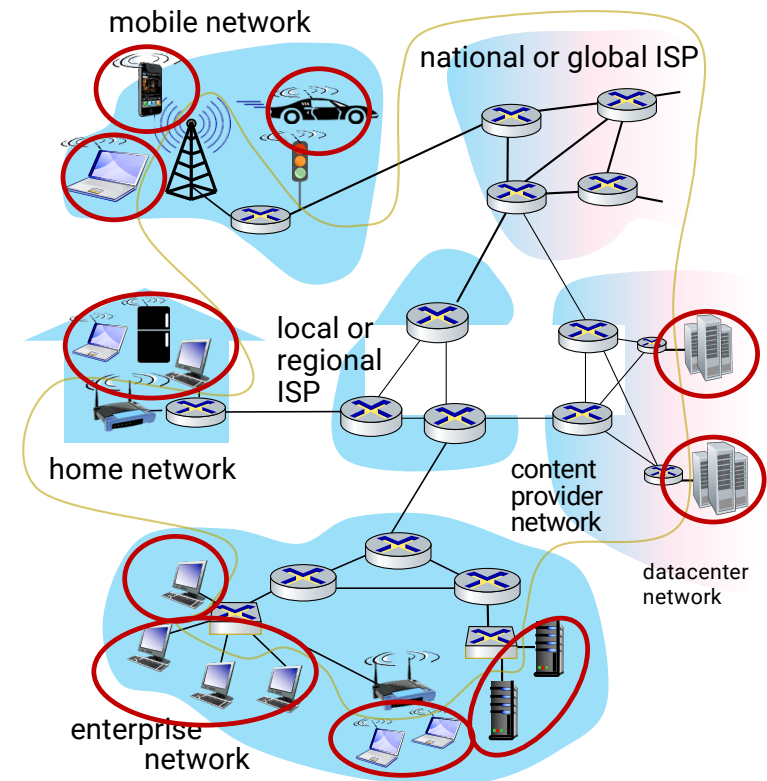
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A closer look at Internet structure

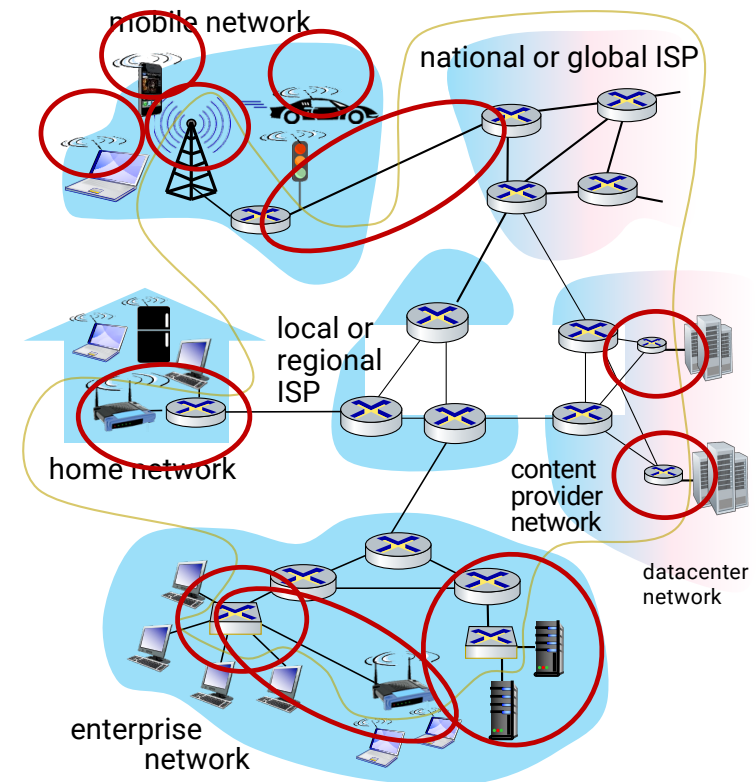
- **Network Edge**

- Hosts: client and servers
- Servers often reside in data centers



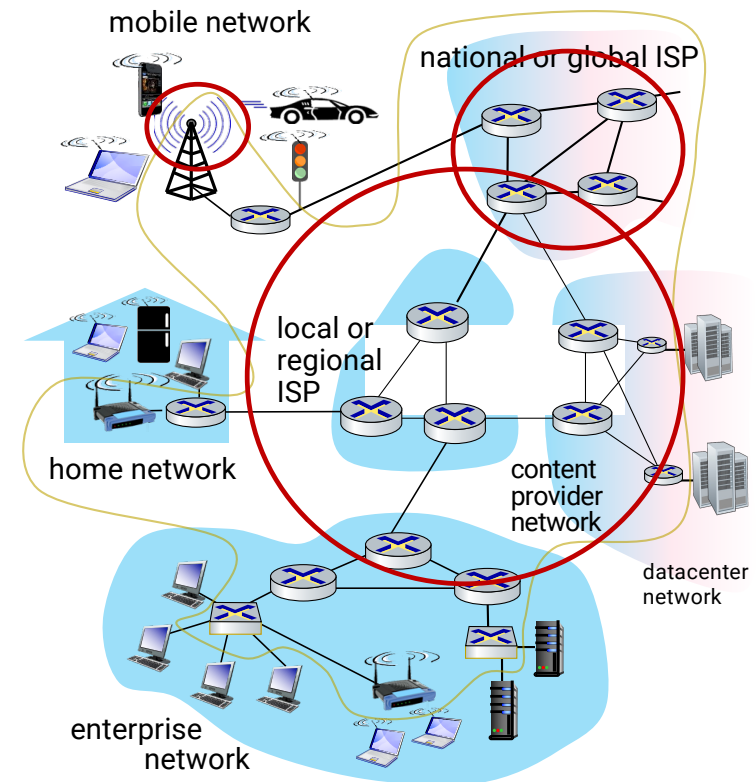
A closer look at Internet structure

- Network Edge
 - Hosts: client and servers
 - Servers often reside in data centers
- Access networks, physical media:
 - Wired links
 - Wireless links



A closer look at Internet structure

- Network Edge
 - Hosts: client and servers
 - Servers often reside in data centers
- Access networks, physical media:
 - Wired links
 - Wireless links
- **Network core:**
 - Interconnected routers
 - Network of networks



Access networks and physical media

Q: How to connect end systems to edge routers?

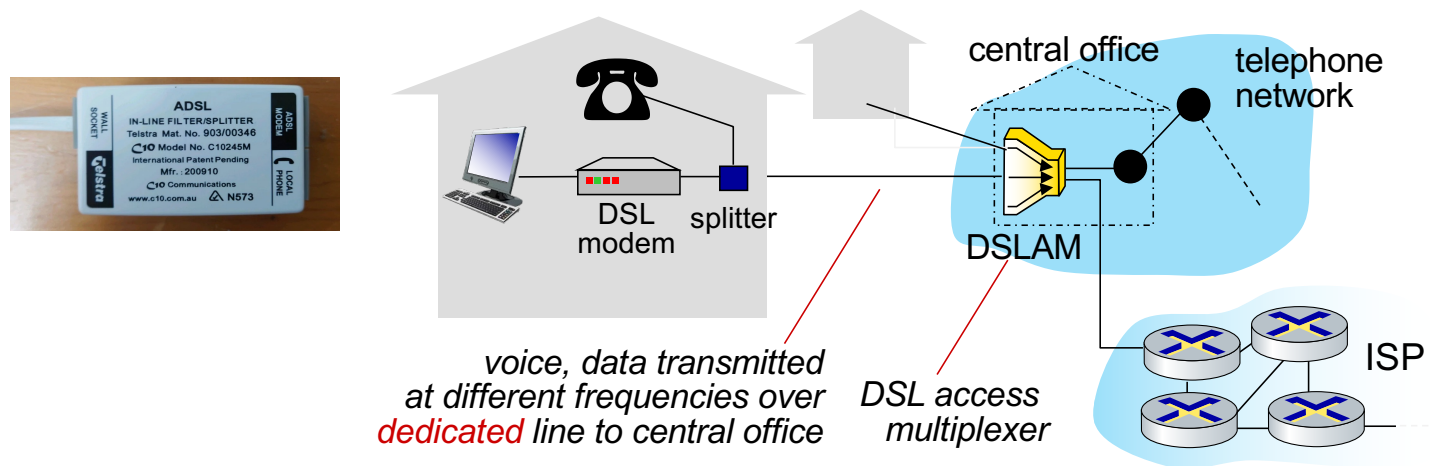
- Residential access nets
- Institutional access networks (school, company)
- Mobile access networks (4G/5G)

What to look for?

Transmission rate (bits per second) of access network?

Shared or dedicated access among users?

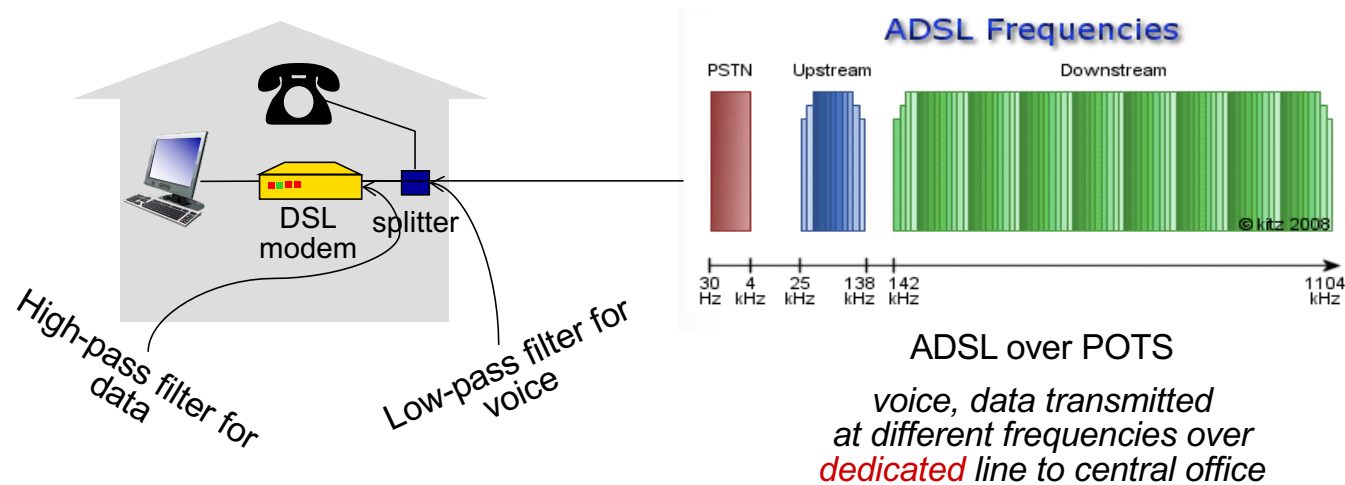
Access networks: digital subscriber line (DSL)



Use **existing** telephone line to central office DSLAM

- data over DSL phone line goes to Internet
- voice over DSL phone line goes to telephone net

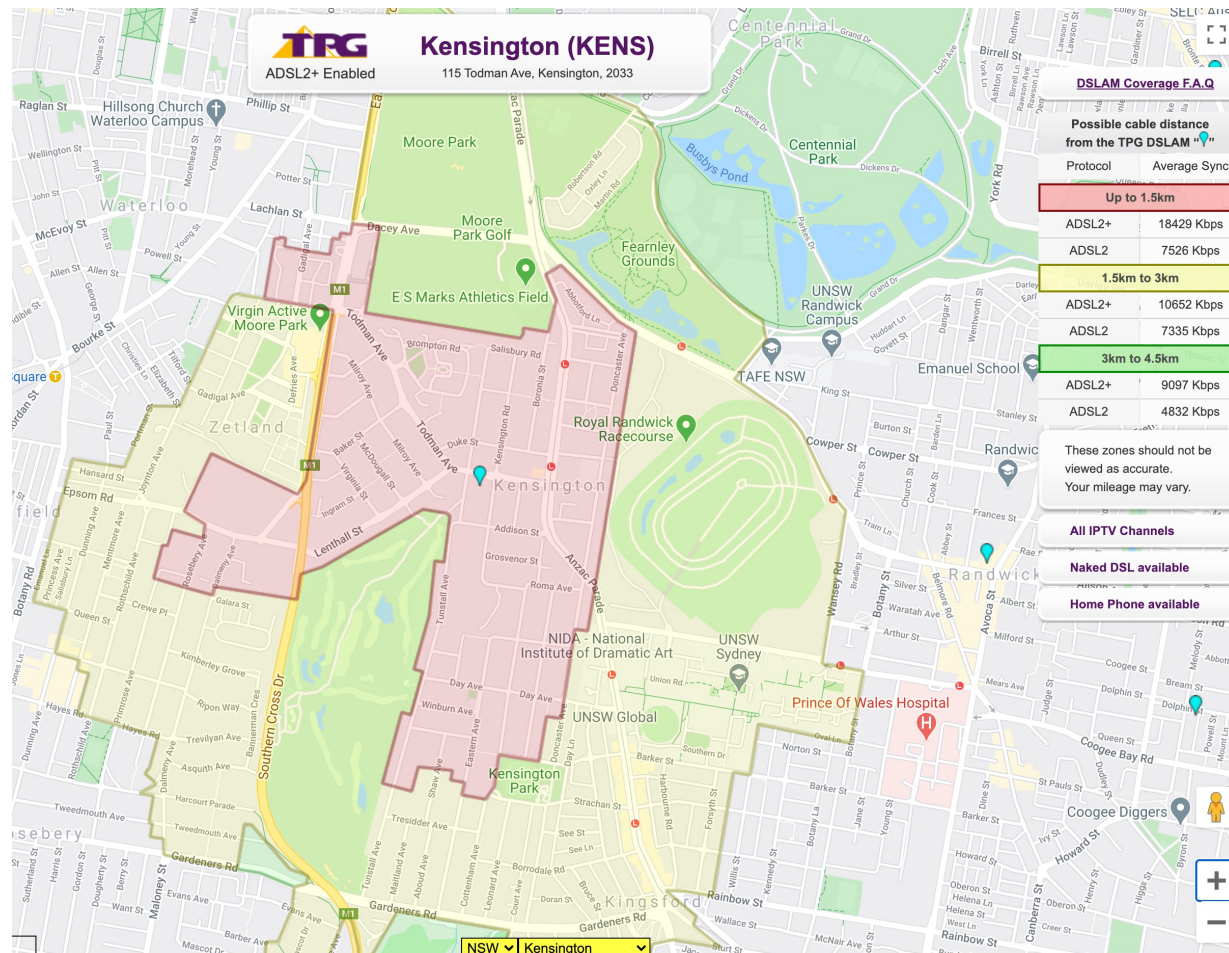
Access networks: digital subscriber line (DSL)



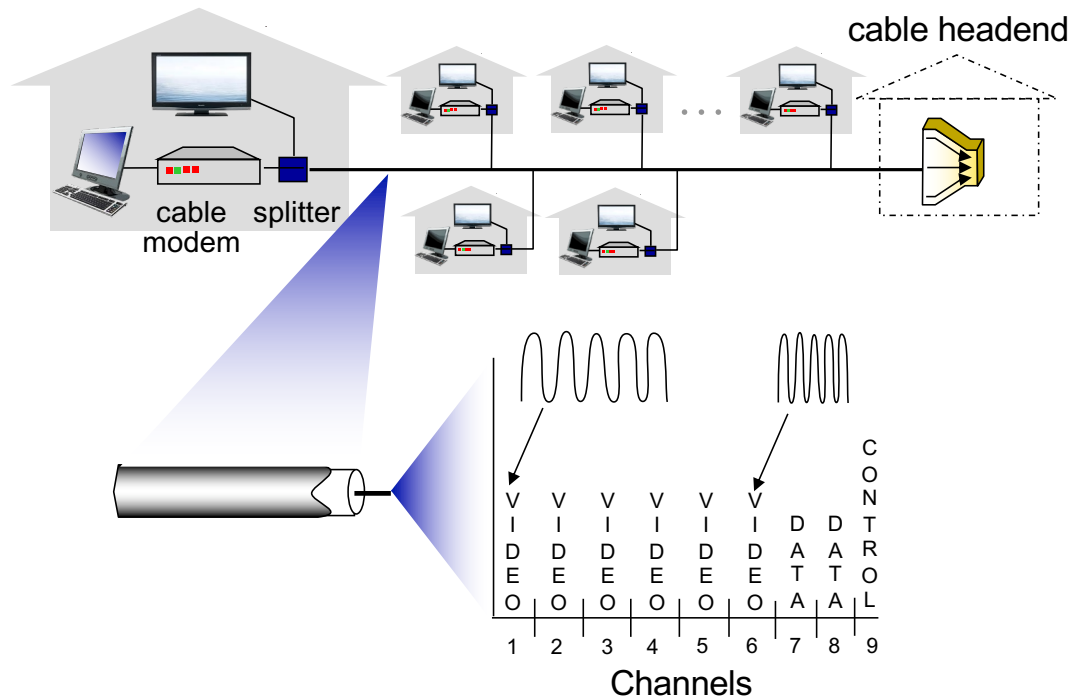
Different data rates for upload and download (ADSL)

- 24-52 Mbps dedicated downstream transmission rate
- 3.5-16 Mbps dedicated upstream transmission rate

Access networks: digital subscriber line (DSL)

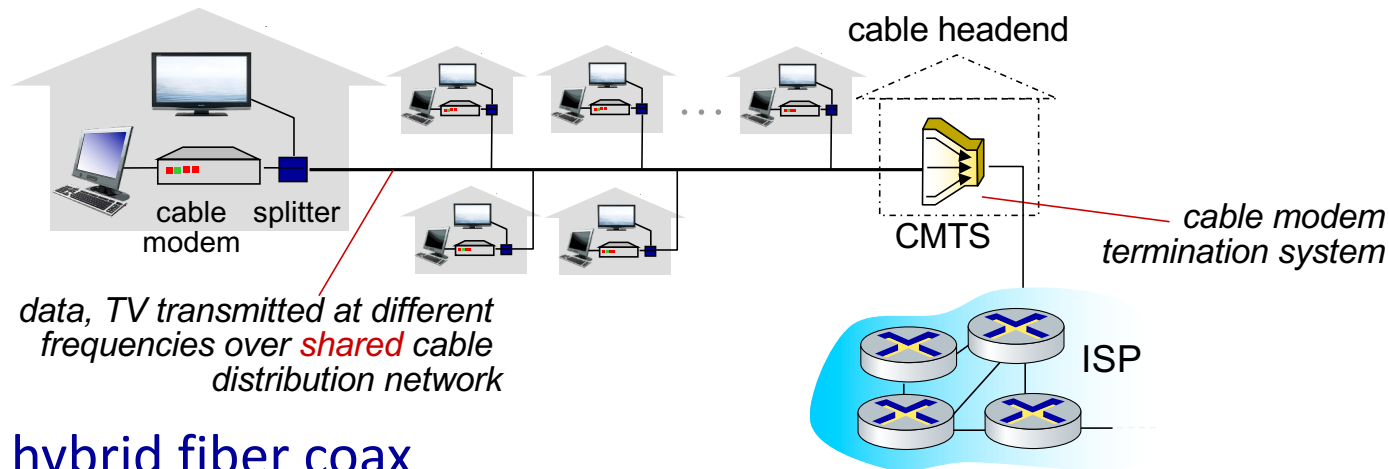


Access networks: cable-based access



frequency division multiplexing (FDM): different channels transmitted in different frequency bands

Access networks: cable-based access



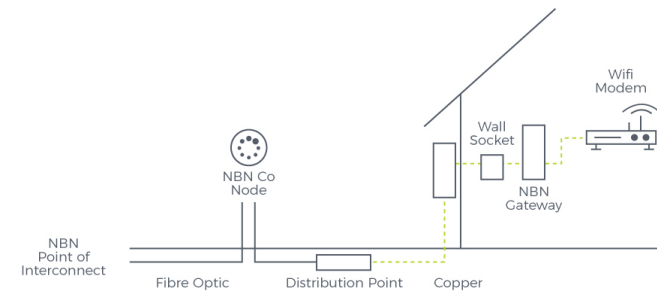
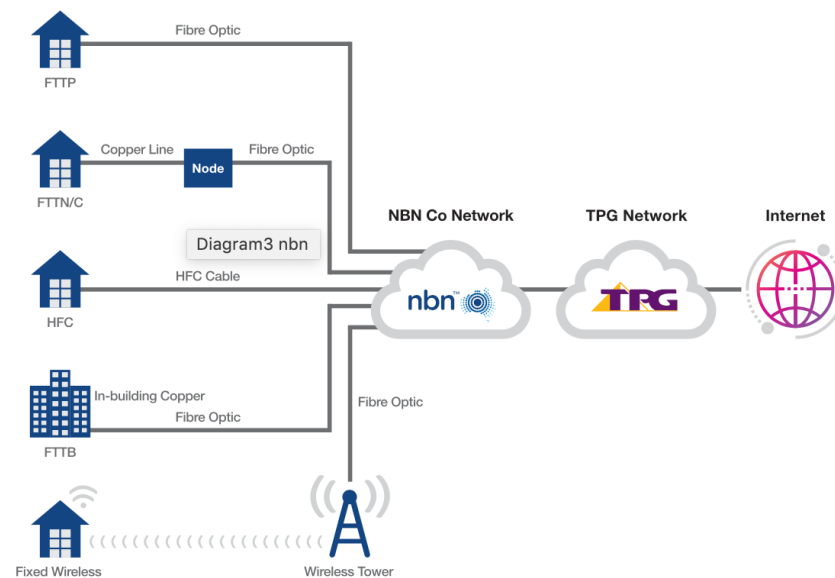
■ HFC: hybrid fiber coax

- asymmetric: up to 40 Mbps – 1.2 Gbps downstream transmission rate, 30-100 Mbps upstream transmission rate
- **network** of cable, fiber attaches homes to ISP router
 - homes *share access network* to cable headend
 - Unlike DSL, which has dedicated access to central office

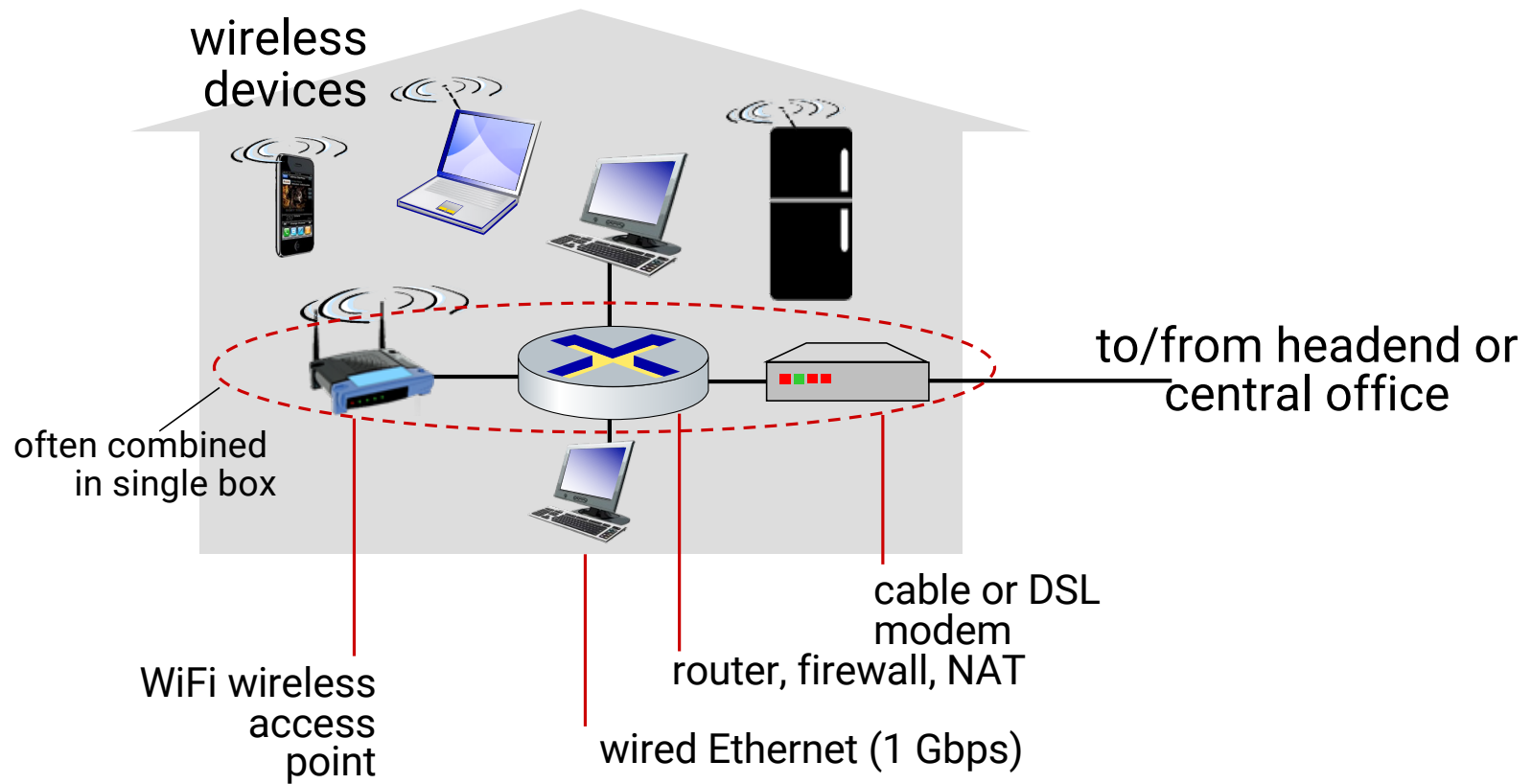
Access networks: FTTP/FTTB/FTTC

Fully optical fiber path all the way to the building (or premise or curb)

- e.g., NBN
- ~30 Mbps to 1Gbps



Access networks: Home networks



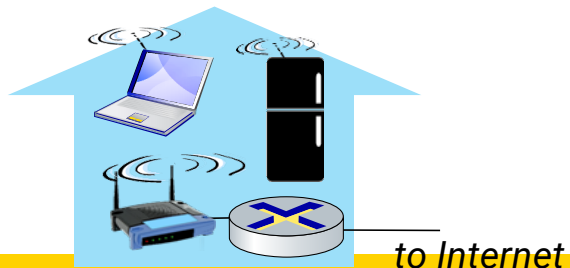
Access networks: Wireless networks

Shared wireless access network connects end system to router

- via base station aka “access point”

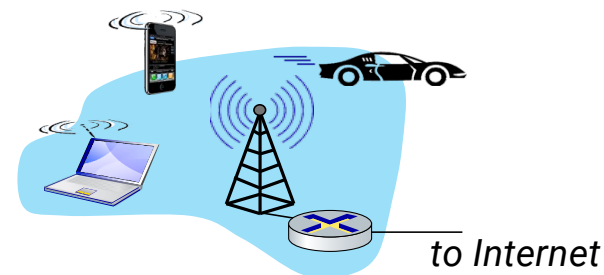
Wireless local area networks (WLANs)

- Typically, within or around building (~100 ft)
- 802.11b/g/n : 11, 54, 450 (Mbps)
- WiFi 5/6/7: 3.5, 9.6, 46 (Gbps)



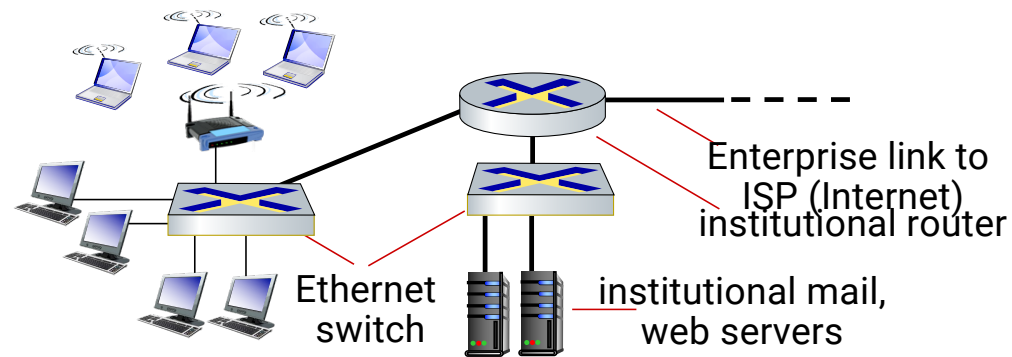
Wide-area cellular access networks

- provided by mobile, cellular network operator (10's km)
- 10's Mbps
- 4G/5G cellular networks



Access networks: Enterprise networks

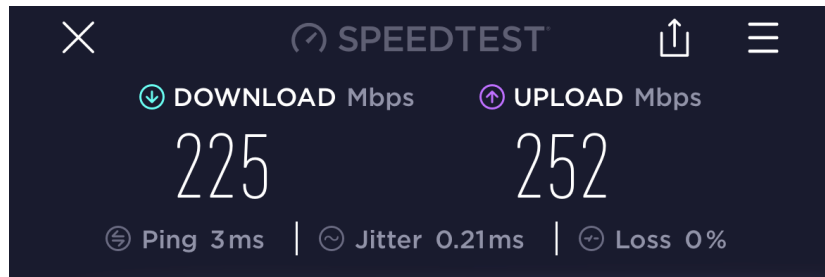
- companies, universities, etc.
- mix of wired, wireless link technologies, connecting a mix of switches and routers (we'll cover differences shortly)
 - Ethernet: wired access at 100Mbps, 1Gbps, 10Gbps



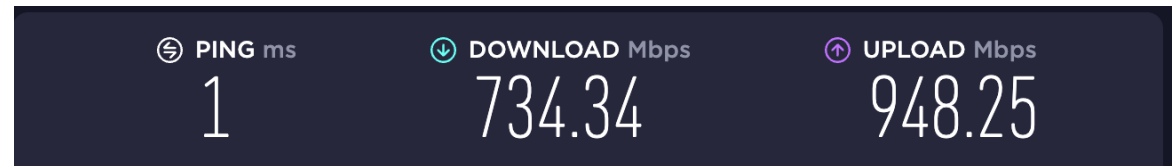
Need for Speed

Can you explain the differences?

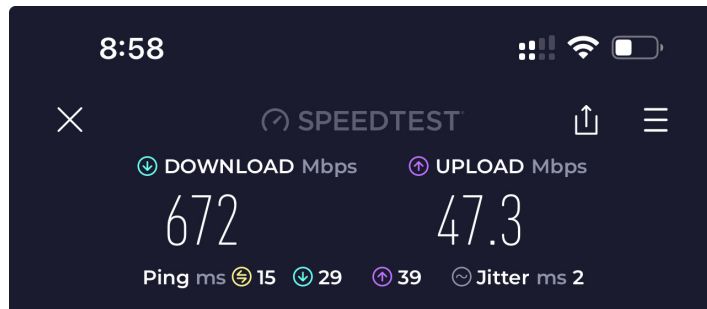
EduRoam



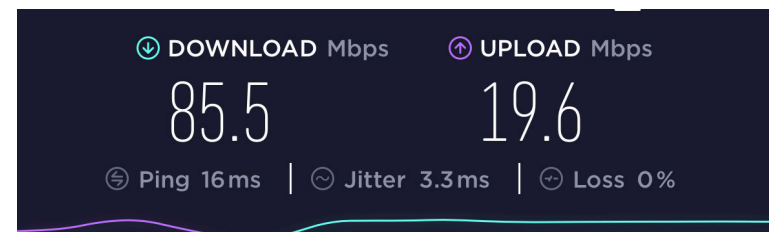
Wired Network @ CSE



NBN (Hybrid Fibre Coaxial or HFC) + WiFi @ my home



4G Network



Quiz 1-3

Which of the following technology is used to connect your home network?

A. FTTB/Copper

B. FTTP

C. FTTC/HFC

D. ADSL

E. Wireless

F. Dial Up



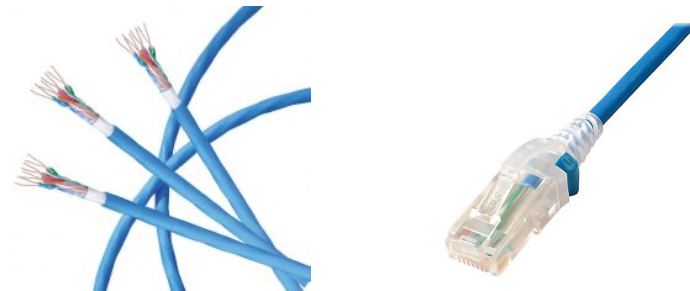
Links: physical media

SELF STUDY
NOT ON EXAM

- **bit**: propagates between transmitter/receiver pairs
- **physical link**: what lies between transmitter & receiver
- **guided media**:
 - signals propagate in solid media: copper, fiber, coax
- **unguided media**:
 - signals propagate freely, e.g., radio

Twisted pair (TP)

- two insulated copper wires
 - Category 5: 100 Mbps, 1 Gbps Ethernet
 - Category 6: 10Gbps Ethernet



Links: physical media

SELF STUDY
NOT ON EXAM

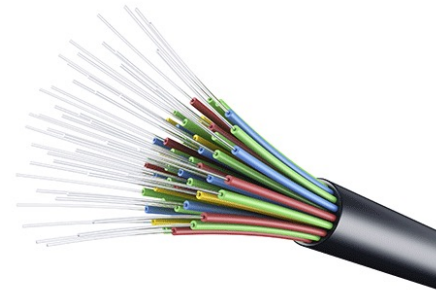
Coaxial cable:

- two concentric copper conductors
- bidirectional
- broadband:
 - multiple frequency channels on cable
 - 100's Mbps per channel



Fiber optic cable:

- glass fiber carrying light pulses, each pulse a bit
- high-speed operation:
 - high-speed point-to-point transmission (10's-100's Gbps)
- low error rate:
 - repeaters spaced far apart
 - immune to electromagnetic noise



Links: physical media

SELF STUDY
NOT ON EXAM

Wireless radio

- signal carried in electromagnetic spectrum
- no physical “wire”
- broadcast and “half-duplex” (sender to receiver)
- propagation environment effects:
 - reflection
 - obstruction by objects
 - interference

Radio link types:

- terrestrial microwave
 - up to 45 Mbps channels
- Wireless LAN (WiFi)
 - Up to 100's Mbps
- wide-area (e.g., cellular)
 - 4G cellular: ~ 10's Mbps
- satellite
 - up to 45 Mbps per channel
 - 270 msec end-end delay
 - geosynchronous versus low-earth-orbit

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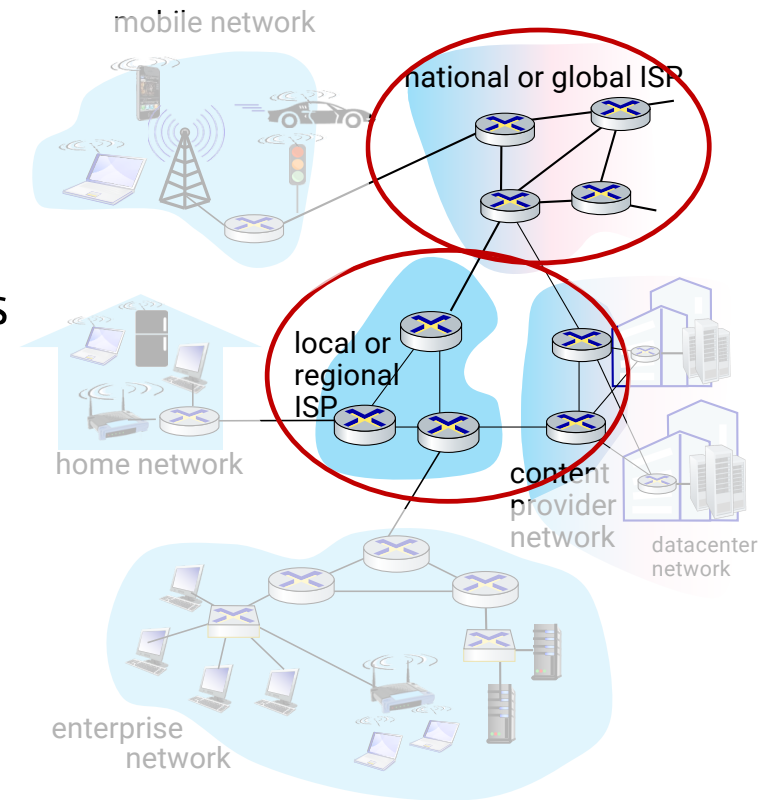
The Network Core

Mesh of interconnected routers

packet-switching: hosts break application-layer messages into packets

- forward packets from one router to the next, across links on path from source to destination
- each packet transmitted at full link capacity
- Is used in the Internet

circuit-switching: an alternative used in legacy telephone networks which was considered during the design of the Internet



Circuit Switching

End-end resources allocated to, reserved for “call” between source and destination

In diagram, each link has four circuits.

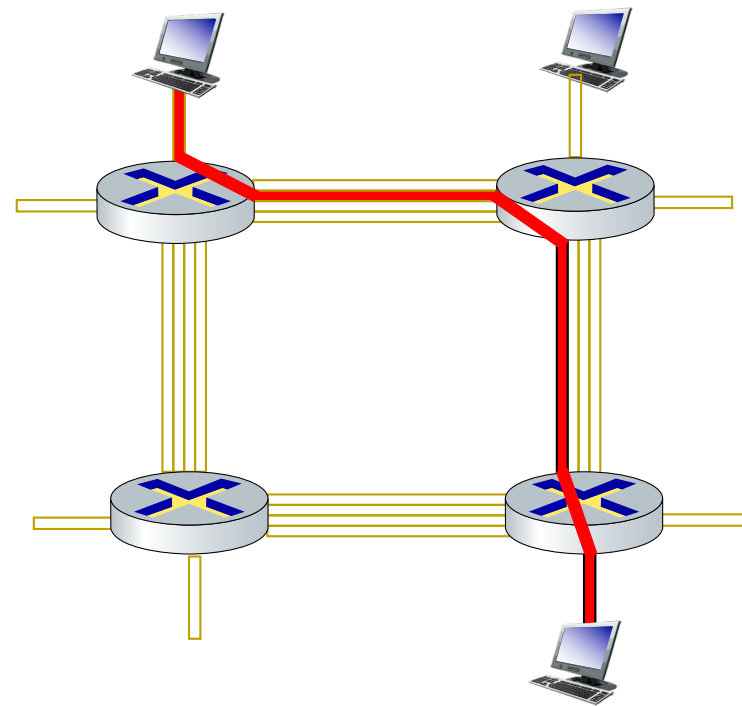
- call gets 2nd circuit in top link and 1st circuit in right link.

Dedicated resources: no sharing

- circuit-like (guaranteed) performance

Circuit segment idle if not used by call (no sharing)

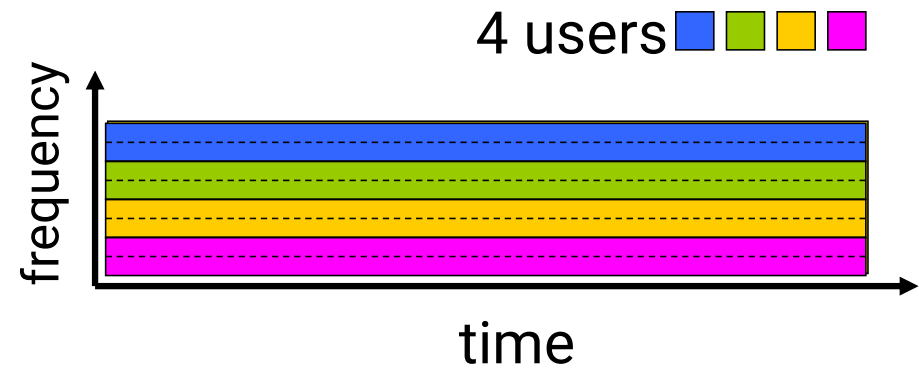
Commonly used in traditional telephone networks



Circuit Switching: FDM and TDM

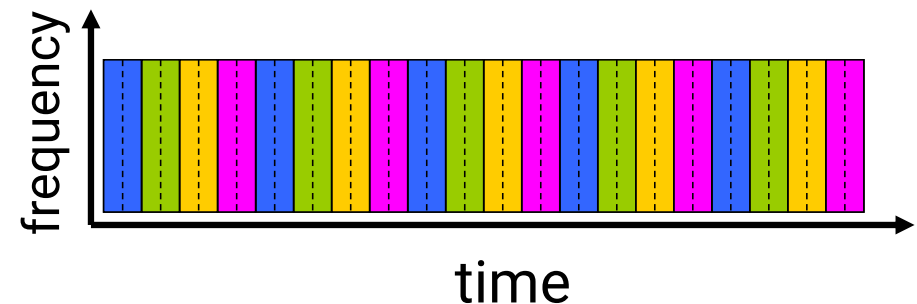
Frequency Division Multiplexing (FDM)

- optical, electromagnetic frequencies divided into (narrow) frequency bands
- each call allocated its own band, can transmit at max rate of that narrow band

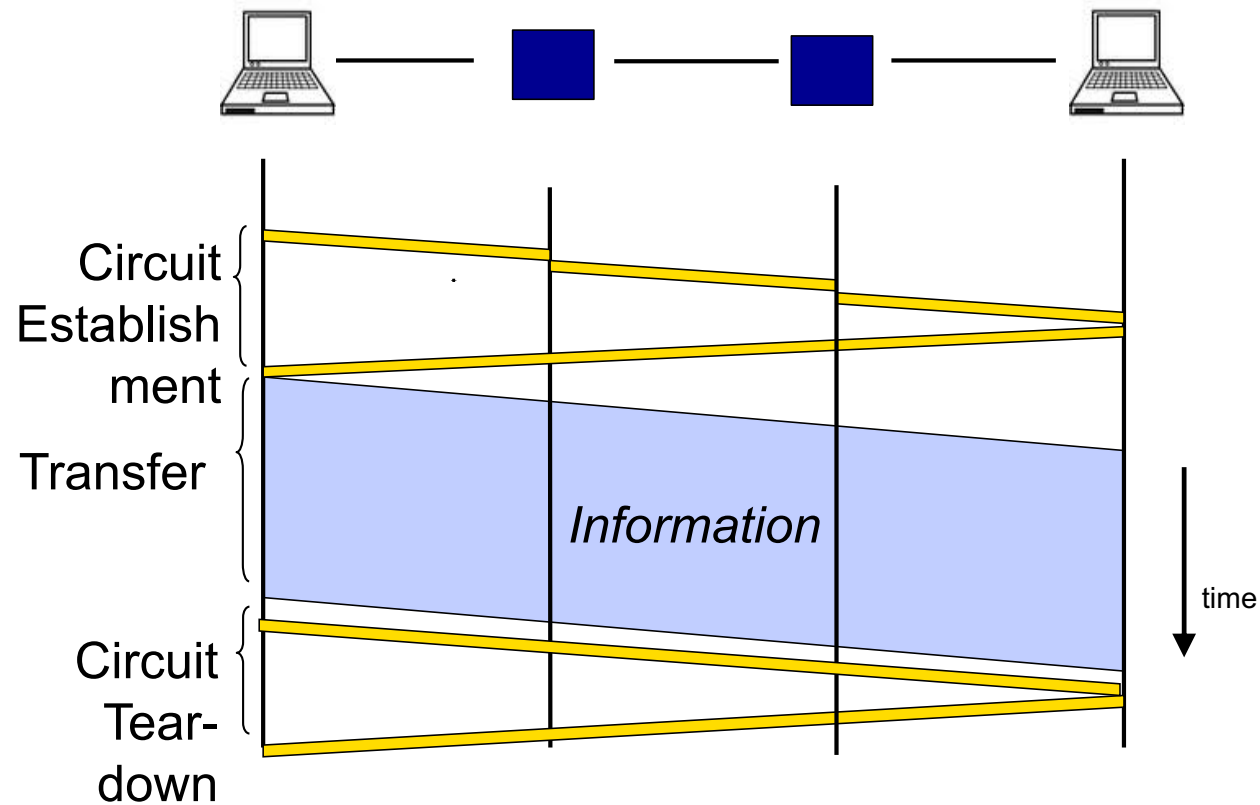


Time Division Multiplexing (TDM)

- time divided into slots
- each call allocated periodic slot(s), can transmit at maximum rate of (wider) frequency band, but only during its time slot(s)



Timing in Circuit Switching

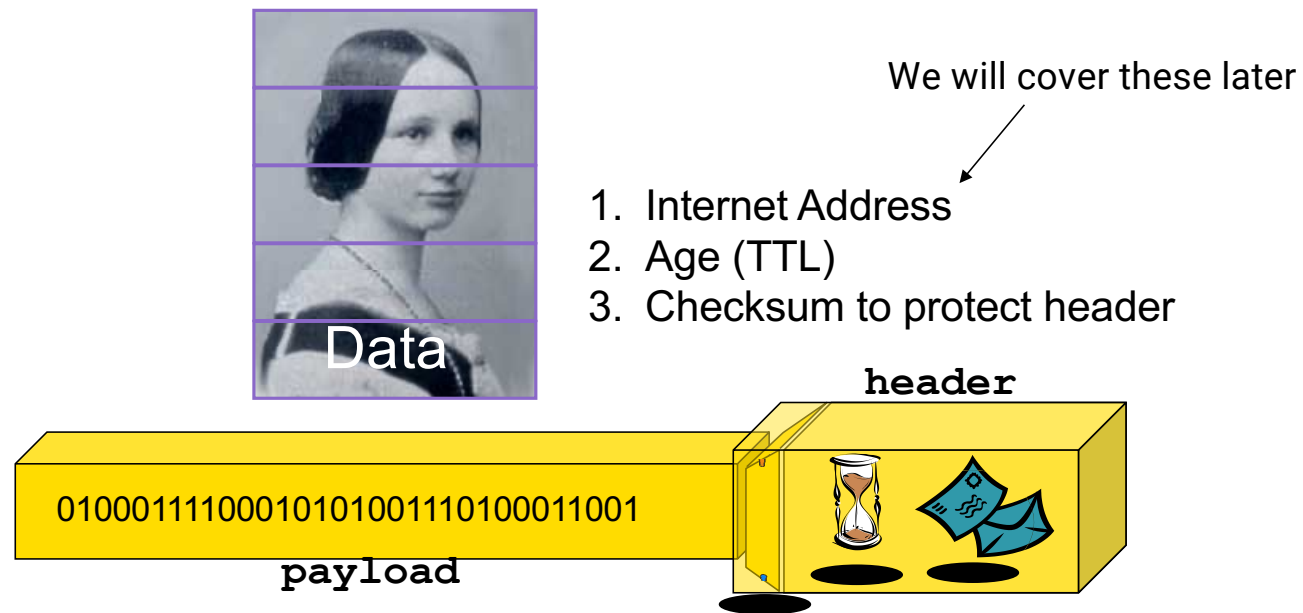


Why Circuit Switching is not feasible?

- **Inefficient**
 - Computer communications tends to be very bursty. For example, viewing a sequence of web pages
 - Dedicated circuit cannot be used or shared in periods of silence
 - Cannot adapt to network dynamics
- **Fixed data rate**
 - Computers communicate at very diverse rates. For example, viewing a video vs using telnet or web browsing
 - Fixed data rate is not useful
- **Connection state maintenance**
 - Requires per communication state to be maintained that is a considerable overhead
 - Not scalable

Packet Switching

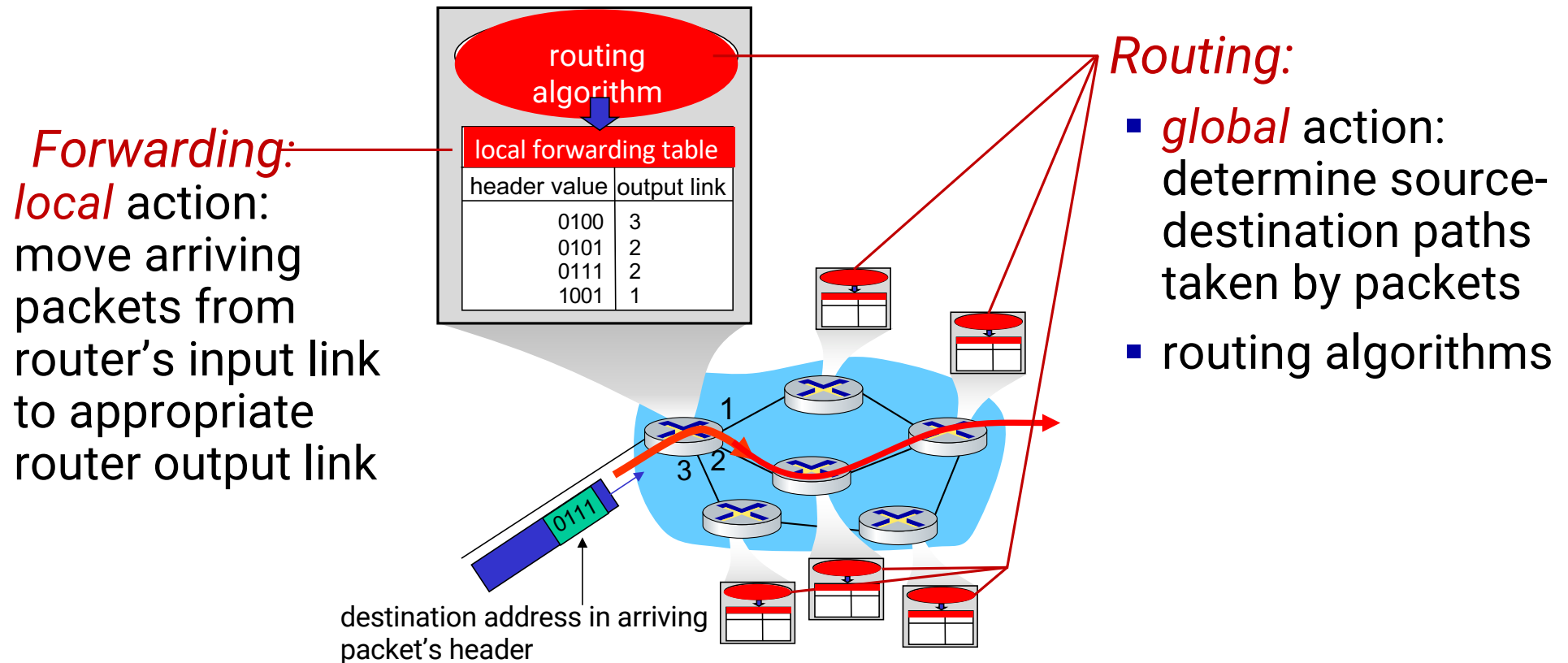
Data is sent as chunks of formatted bits (Packets)
Packets consist of a "header" and "payload"



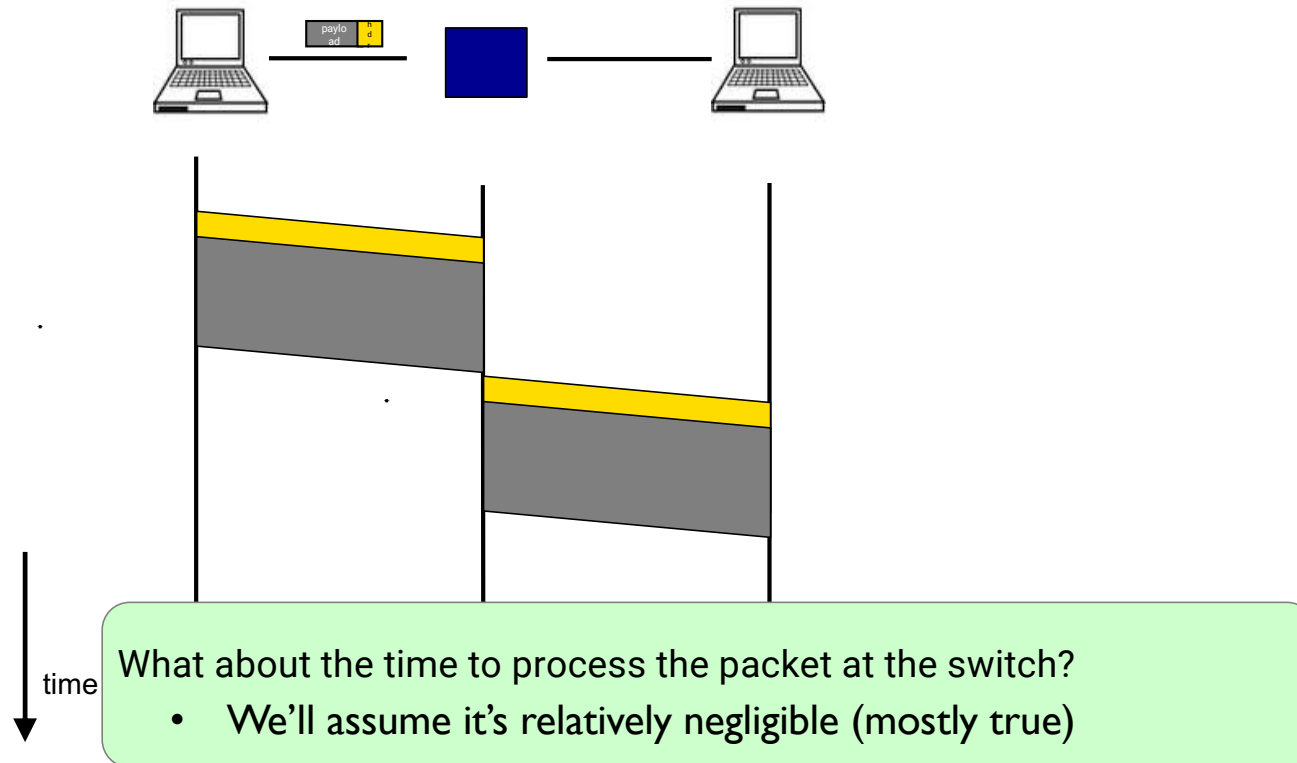
Packet Switching

- Data is sent as chunks of formatted bits (**Packets**)
- Packets consist of a "**header**" and "**payload**"
 - payload is the data being carried
 - header holds instructions to the network for how to handle packet (think of the header as an API)
- Switches/routers "**forward**" packets based on their headers

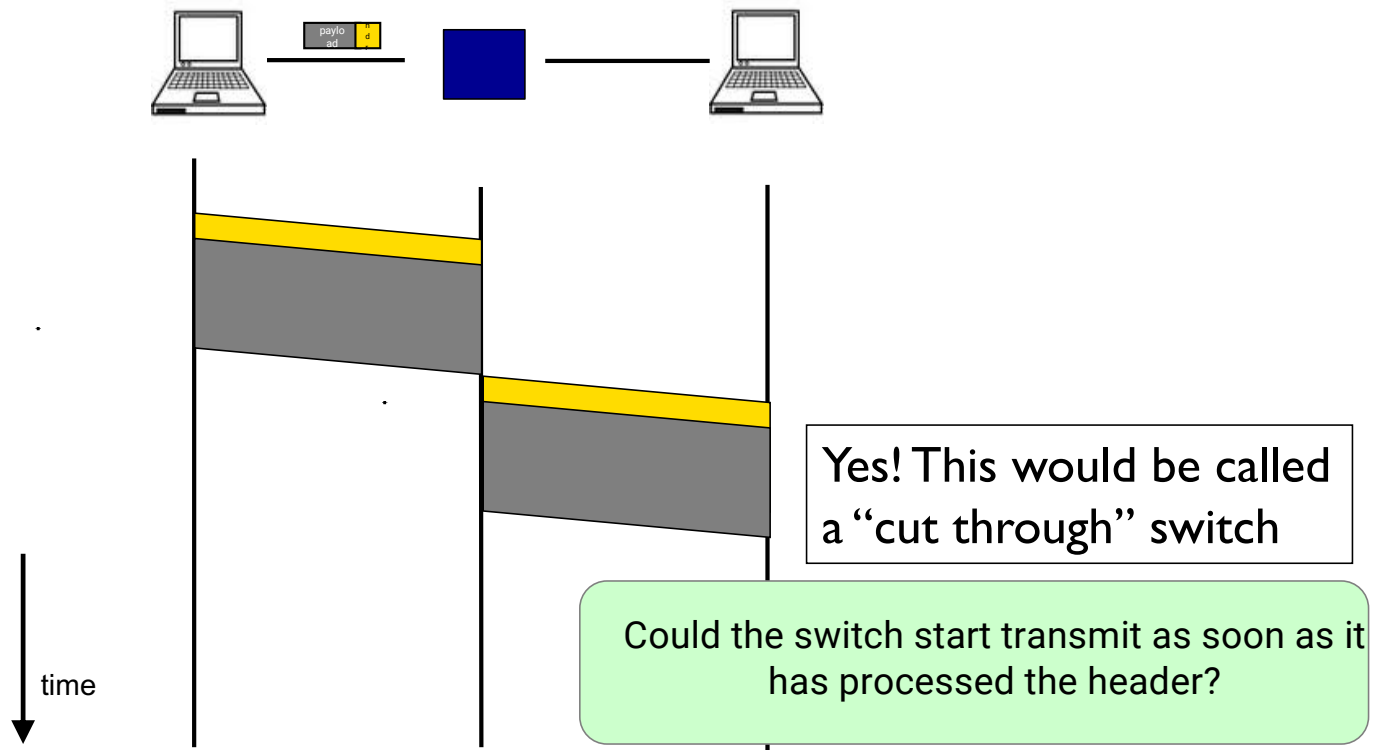
Peek ahead: Two key network-core functions



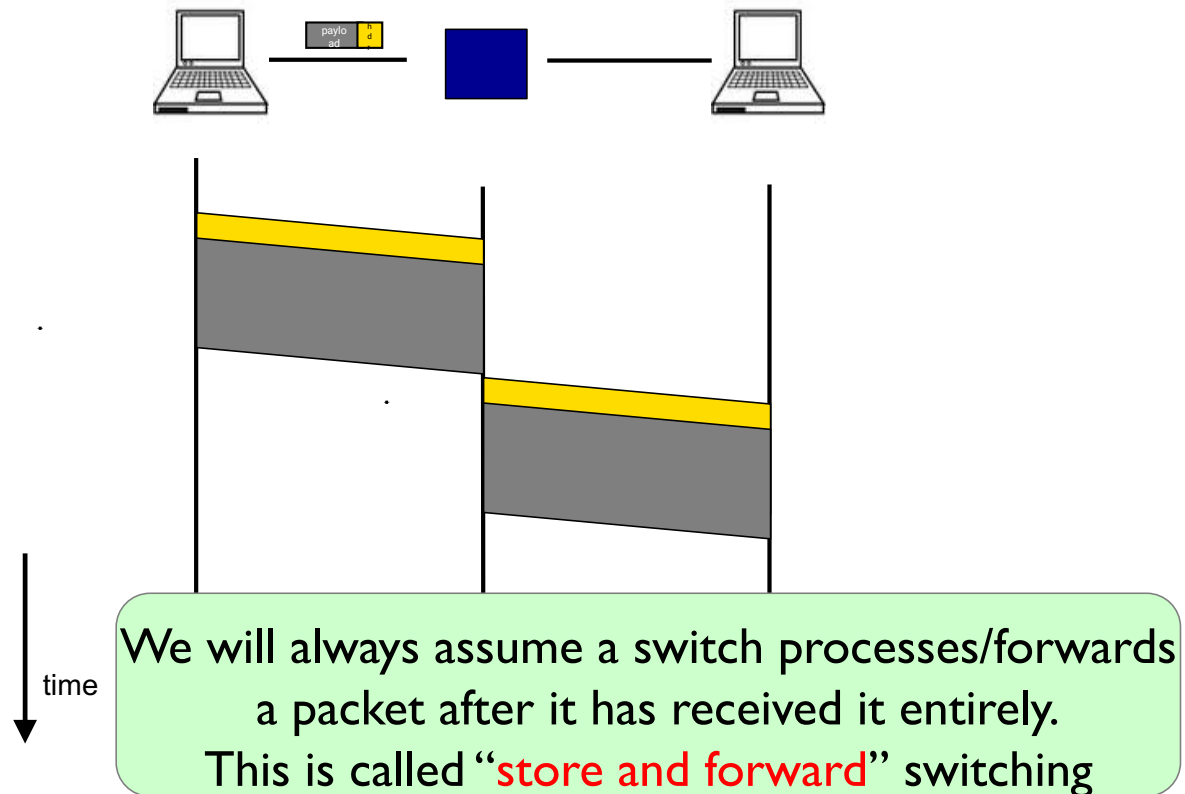
Timing in Packet Switching



Timing in Packet Switching



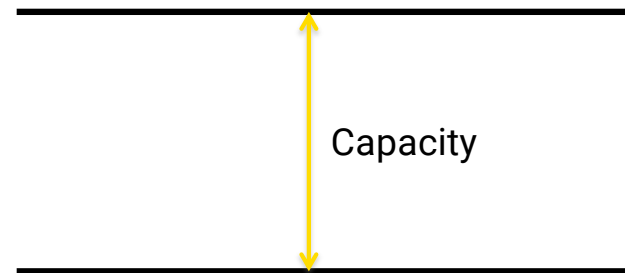
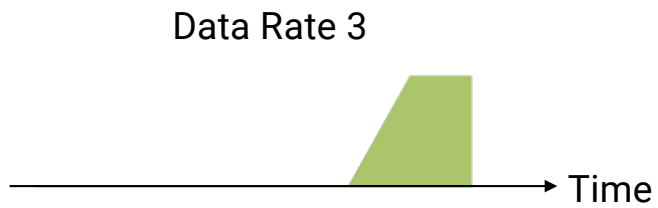
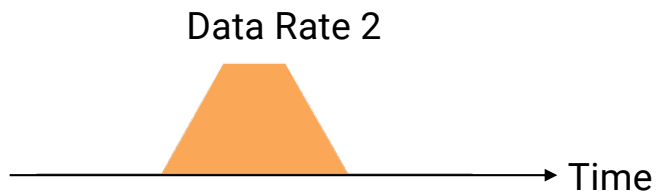
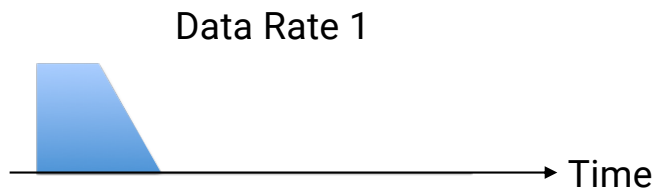
Timing in Packet Switching



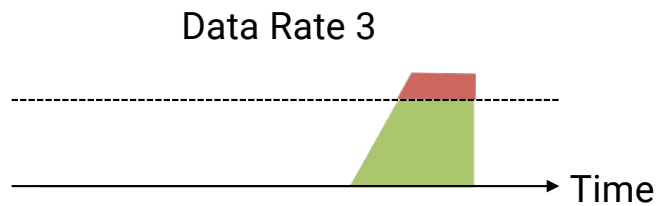
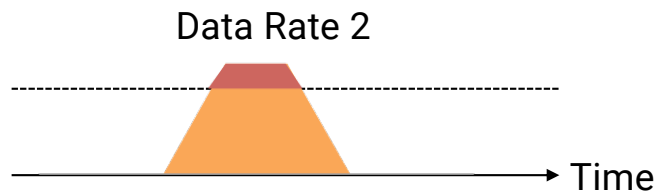
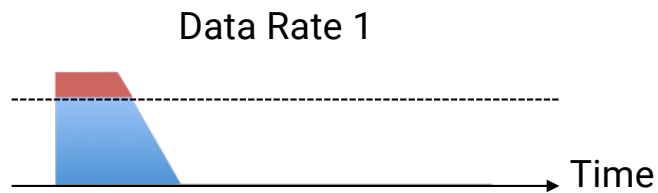
Packet Switching

- Data is sent as chunks of formatted bits (**Packets**)
 - Packets consist of a "**header**" and "**payload**"
 - payload is the data being carried
 - header holds instructions to the network for how to handle packet (think of the header as an API)
 - Switches/routers "**forward**" packets based on their headers
- Each packet travels independently
 - no notion of packets belonging to a "circuit"
 - No link resources are reserved. Instead, packet switching leverages **statistical multiplexing**

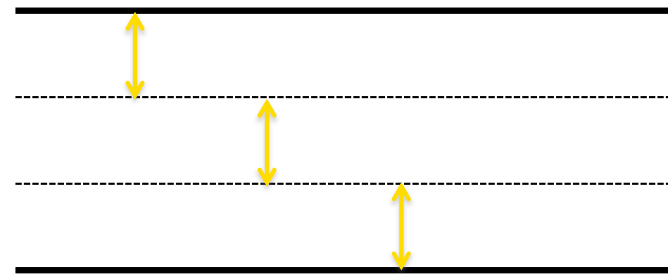
Three Flows with Bursty Traffic



When Each Flow Gets 1/3rd of Capacity



like circuit switching



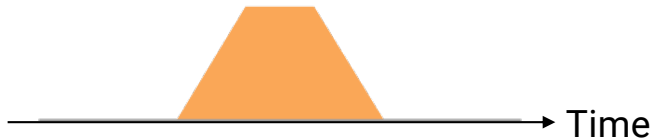
Overloaded

When Flows Share Total Capacity

packet switching



No Overloading

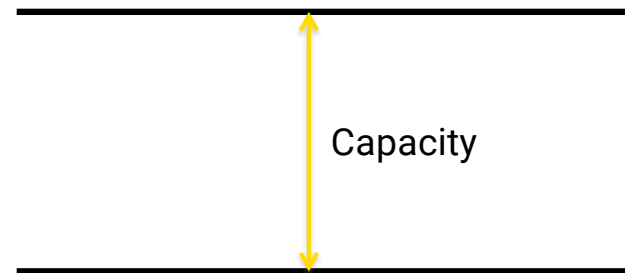
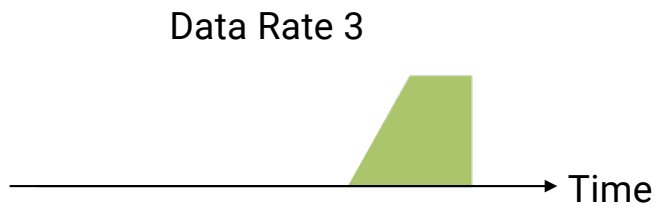
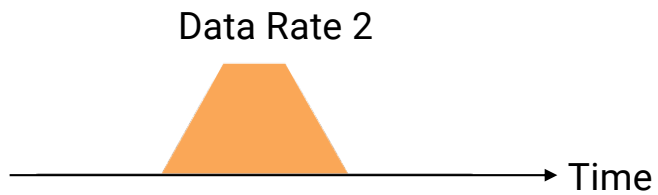
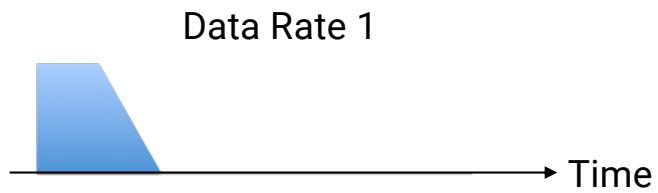


Statistical multiplexing relies on the assumption that not all flows burst at the same time

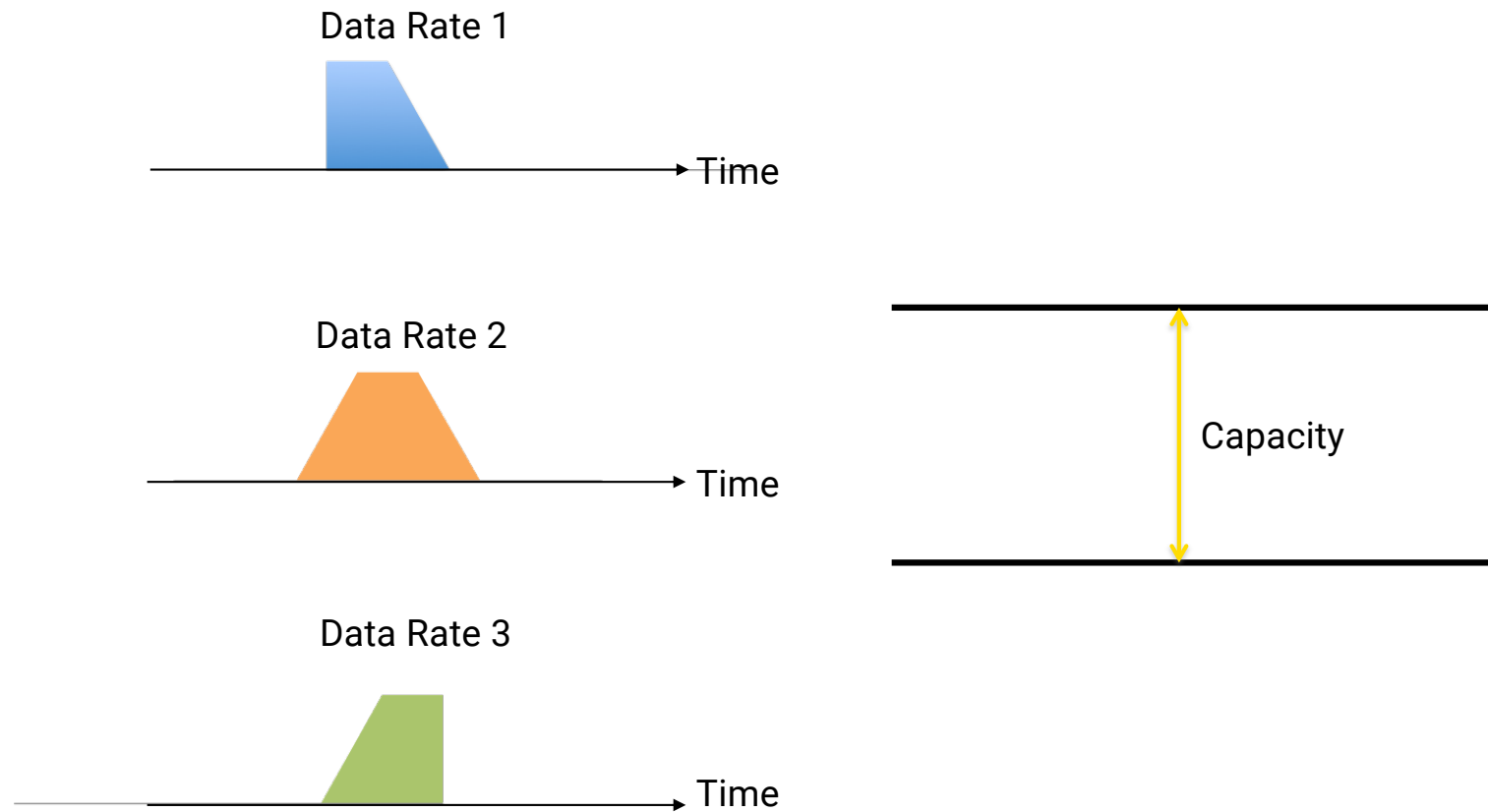


Very similar to insurance, and has same failure case

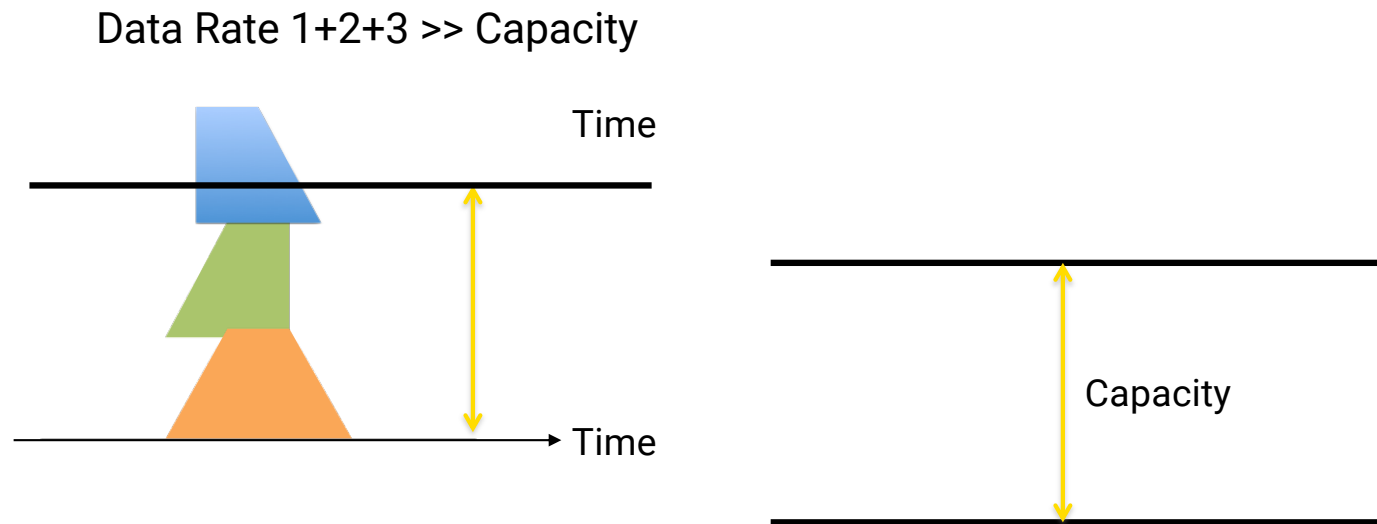
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Three Flows with Bursty Traffic

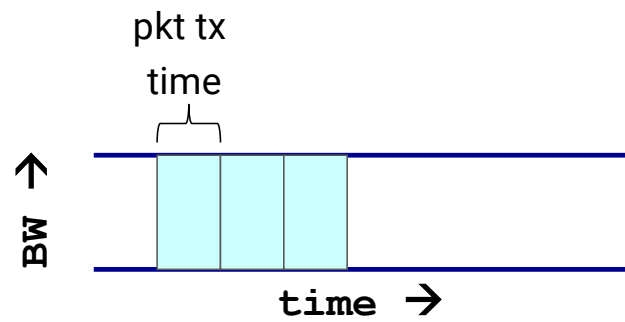


Three Flows with Bursty Traffic

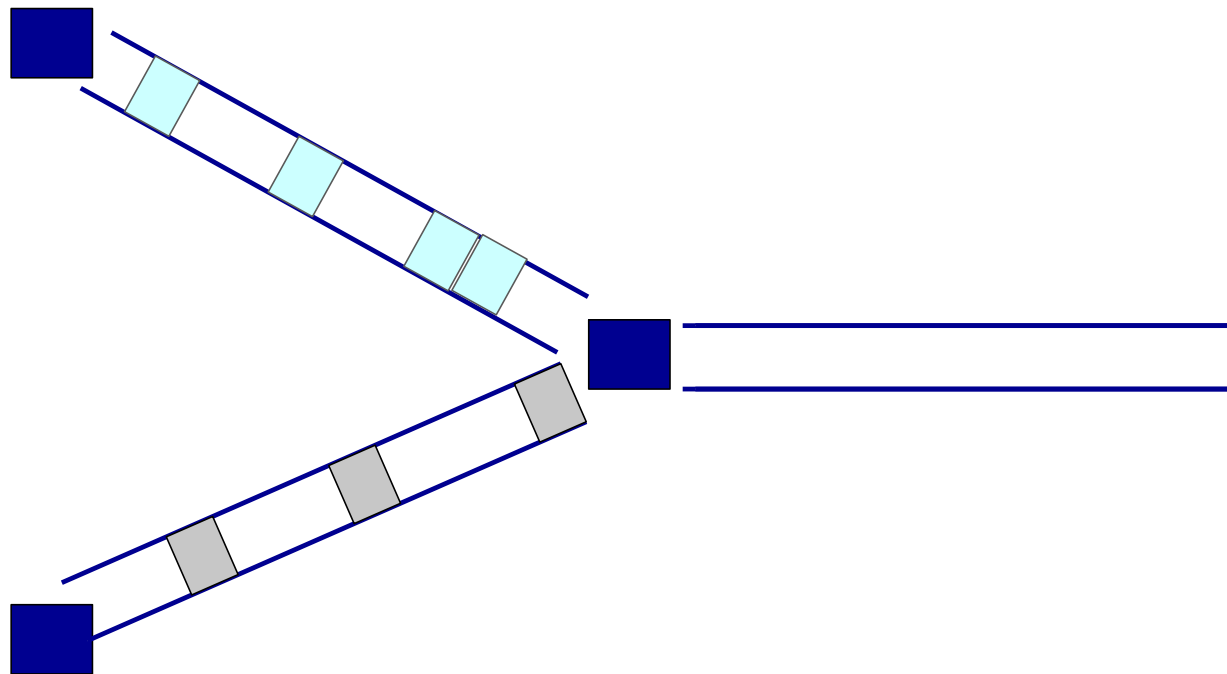


What do we do under overload?

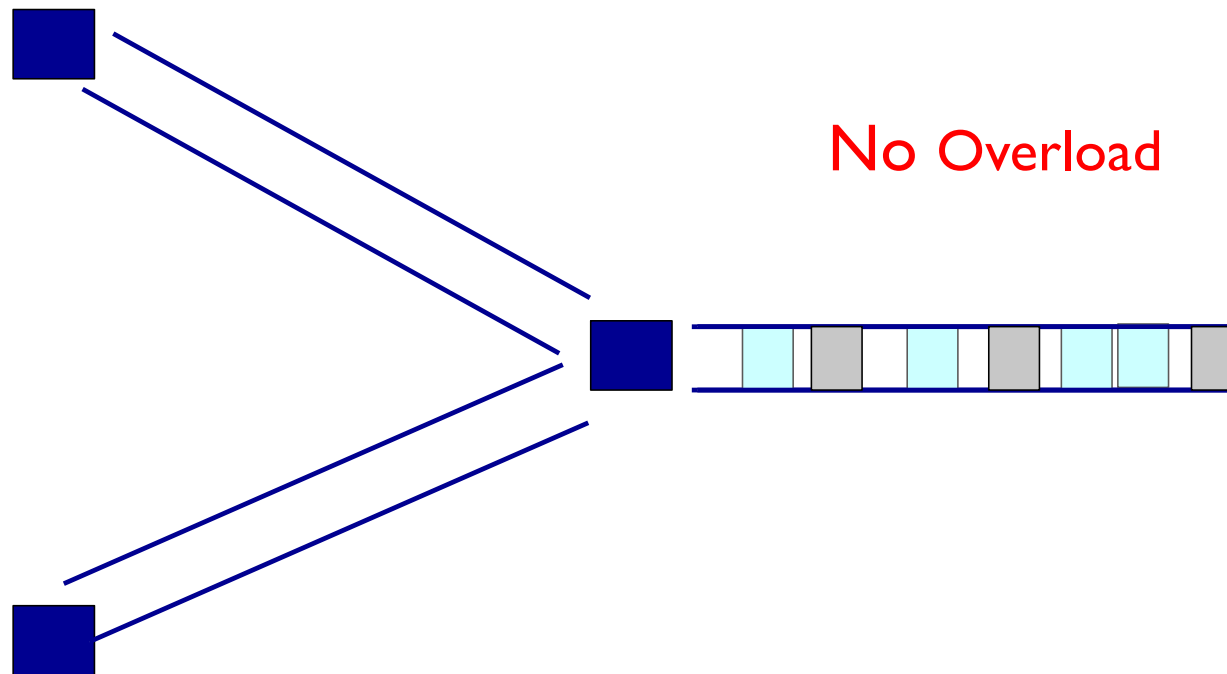
Statistical multiplexing: pipe view



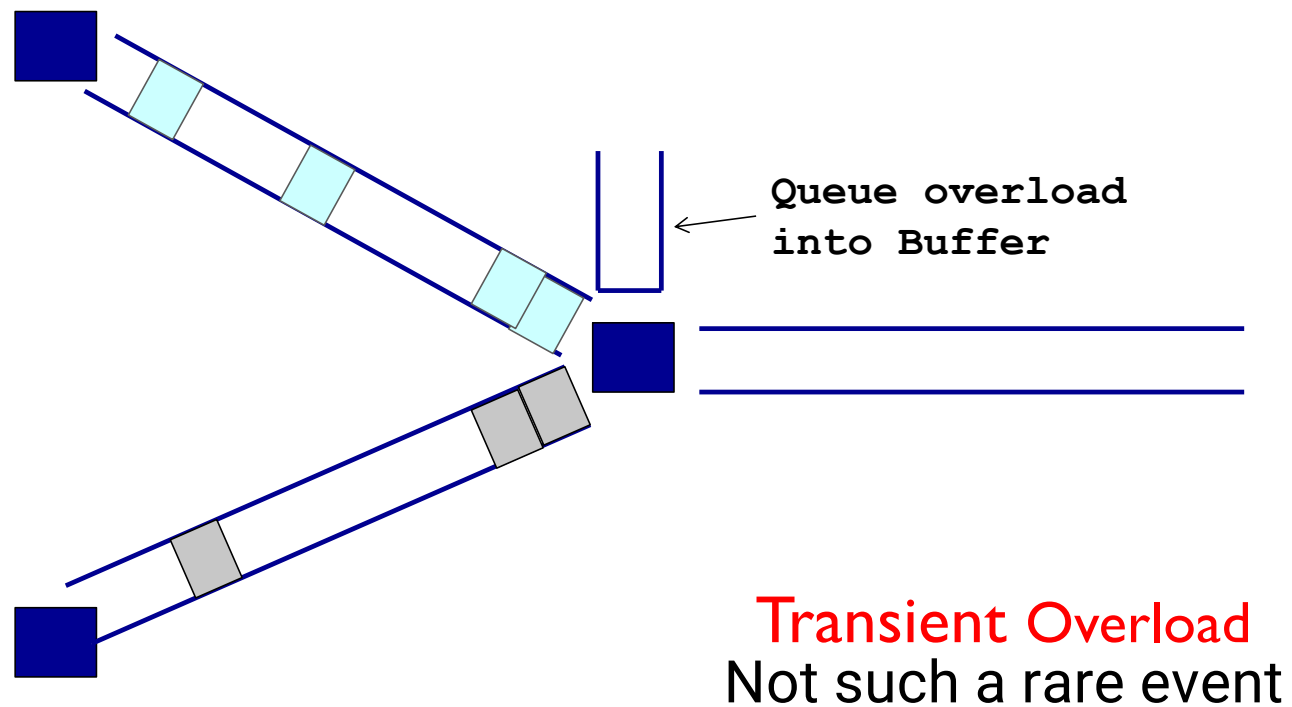
Statistical multiplexing: pipe view



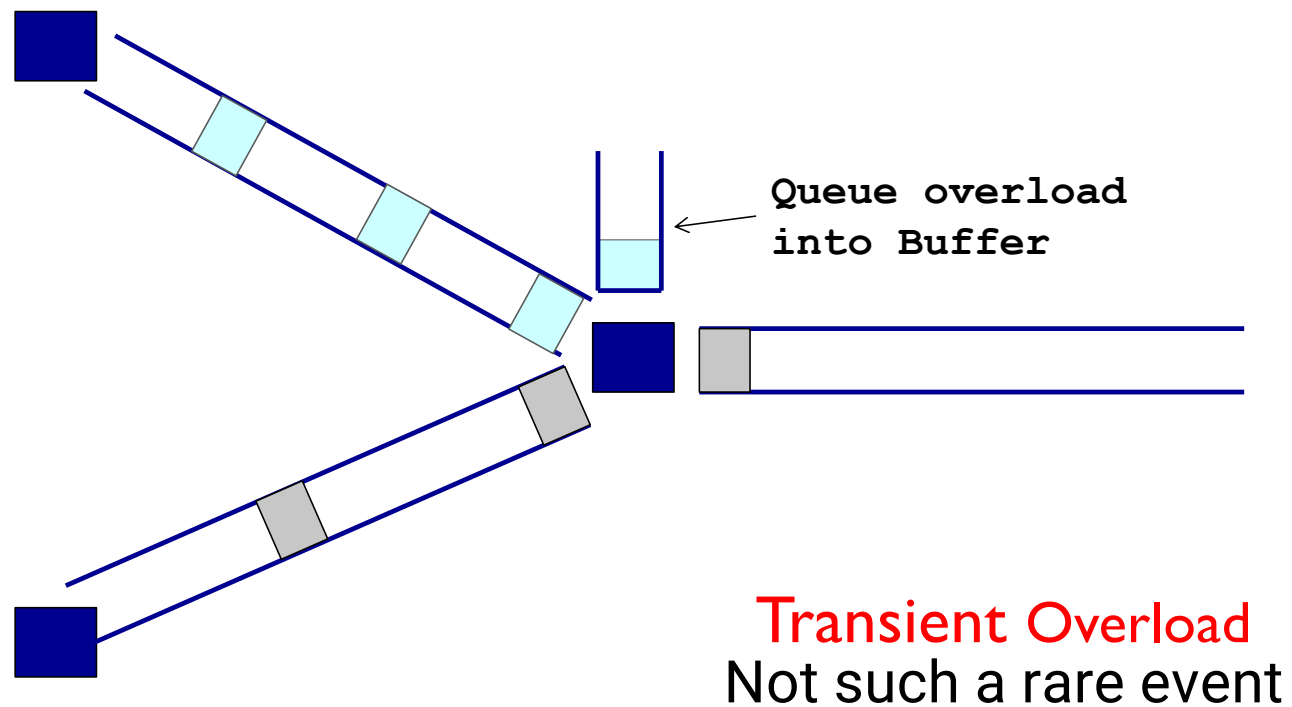
Statistical multiplexing: pipe view



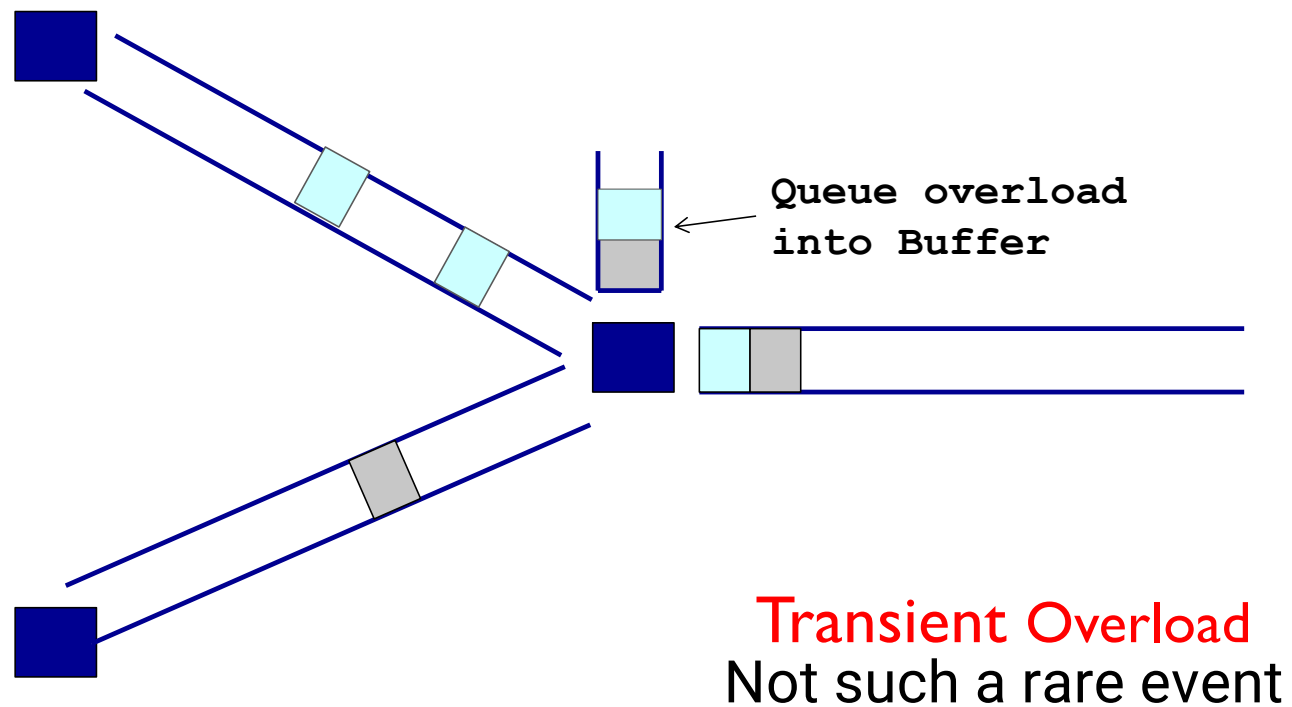
Statistical multiplexing: pipe view



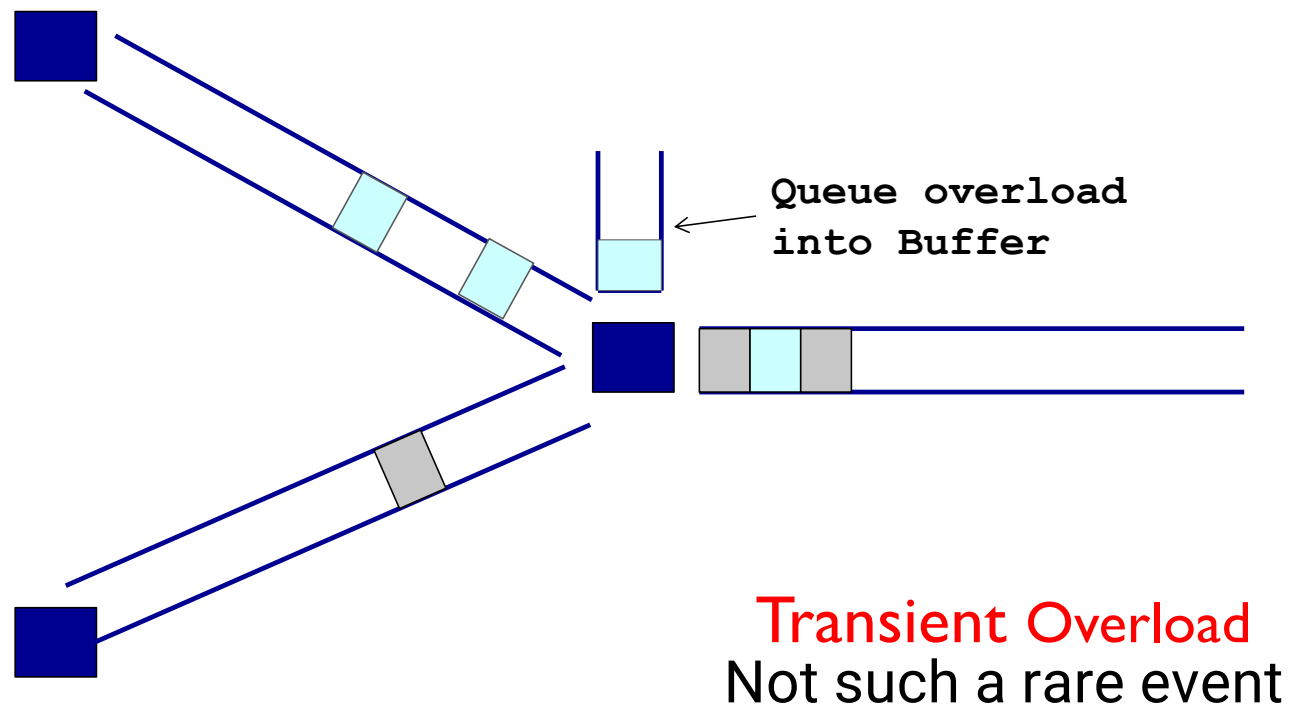
Statistical multiplexing: pipe view



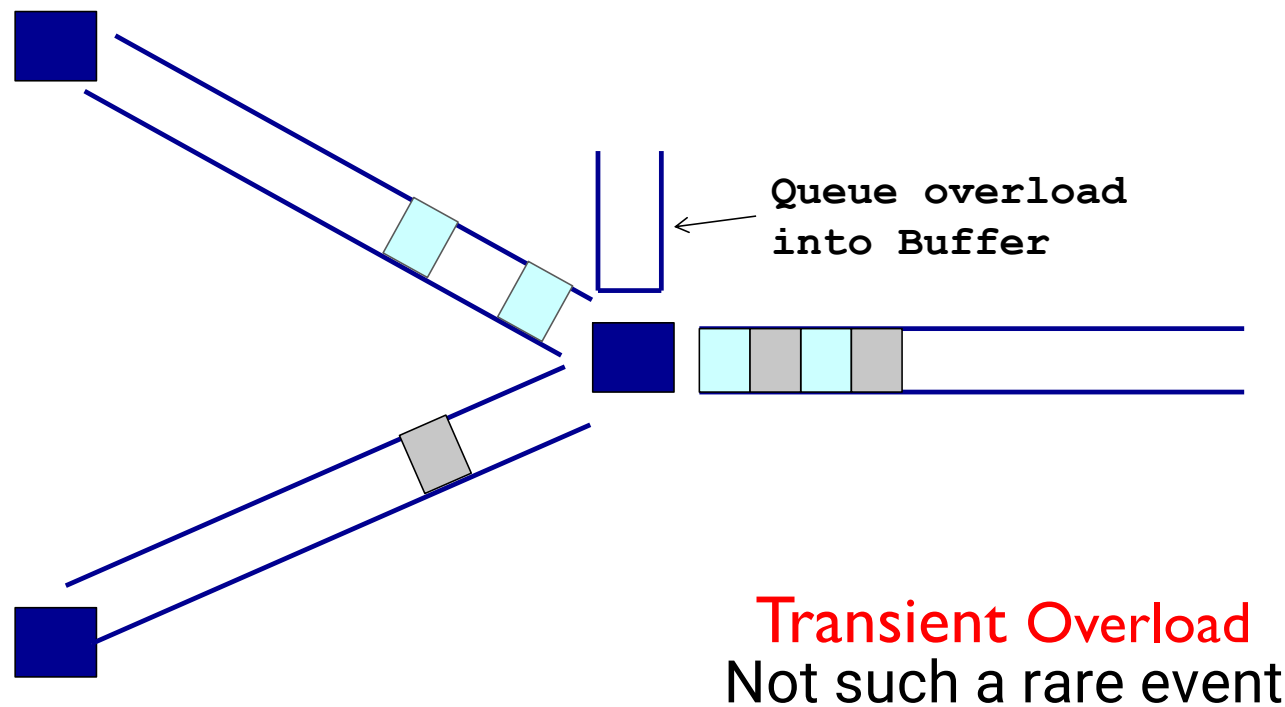
Statistical multiplexing: pipe view



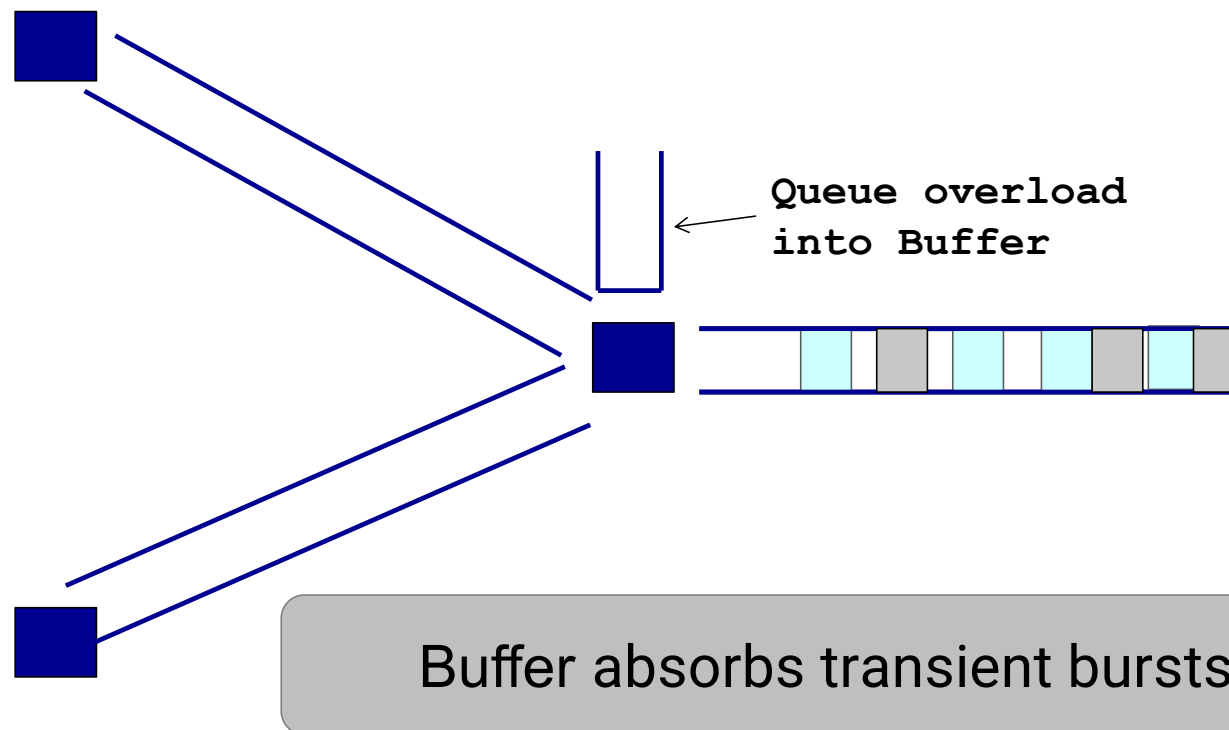
Statistical multiplexing: pipe view



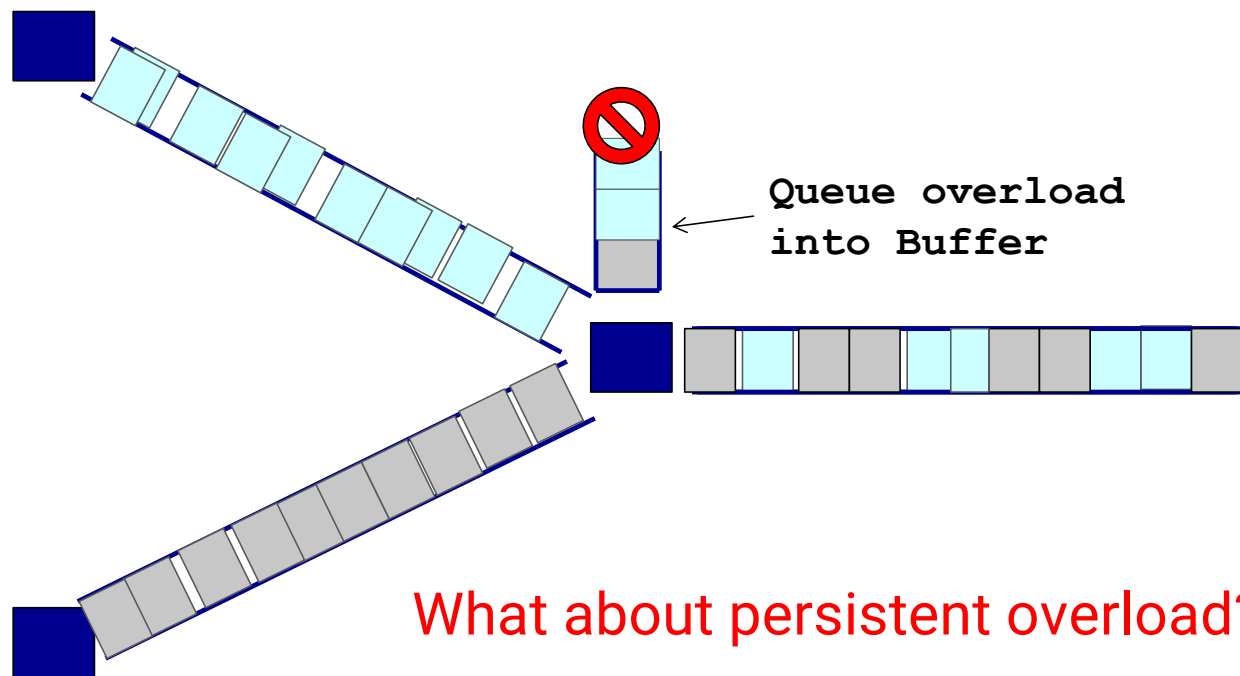
Statistical multiplexing: pipe view



Statistical multiplexing: pipe view



Statistical multiplexing: pipe view



What about persistent overload?

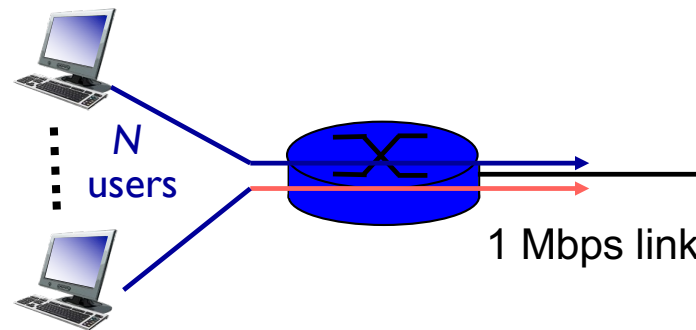
Will eventually drop packets

Packet Switching vs Circuit Switching

Packet switching allows more users to use network!

Example:

- 1 Mb/s link
- each user:
 - 100 kb/s when “active”
 - active 10% of time



Circuit-switching:

- 10 users

Packet switching:

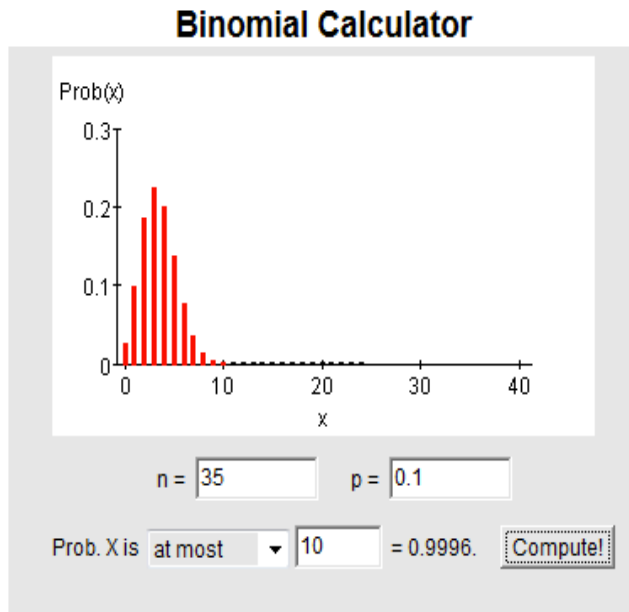
- with 35 users, probability > 10 active at same time is less than .0004

Q: how did we get value 0.0004?

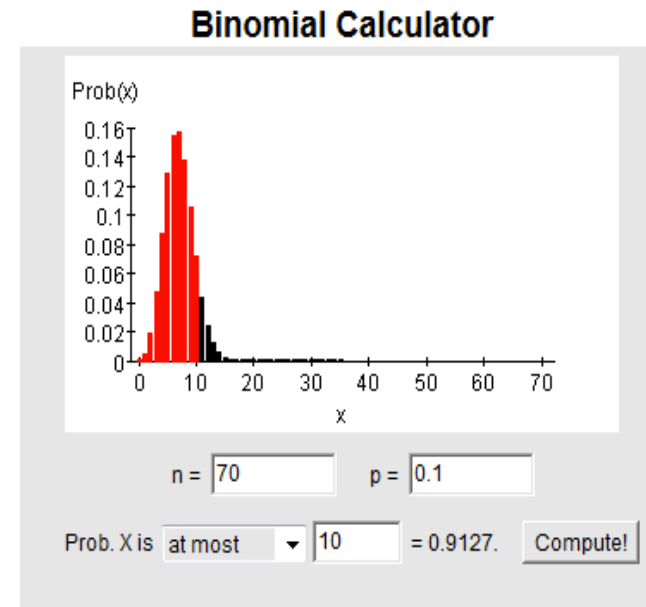
Q: what happens if > 35 users say 70?

Hint: Bernoulli Trials and Binomial Distribution

Statistical Multiplexing Gain (SMG)



SMG: 35/10=3.5



SMG: 70/10=7

Packet Switching vs Circuit Switching

Is packet switching a “slam dunk winner”?

Great for “bursty” data – sometimes has data to send, but at other times not

- resource sharing
- simpler, no call setup

Excessive congestion possible: packet delay and loss due to buffer overflow

- protocols needed for reliable data transfer, congestion control

Q: How to provide circuit-like behavior?

- bandwidth guarantees traditionally used for audio/video applications

Q: human analogies of reserved resources (circuit switching)
versus on-demand allocation (packet switching)?

Quiz 1-4

In _____ resources are allocated on demand

- A. Packet switching
- B. Circuit switching
- C. Both
- D. None



Quiz 1-5

A message from device A to B consists of packet X and packet Y. In a circuit switched network, packet Y's path _____ packet X's path

- A. is the same
- B. is independent
- C. is always different from

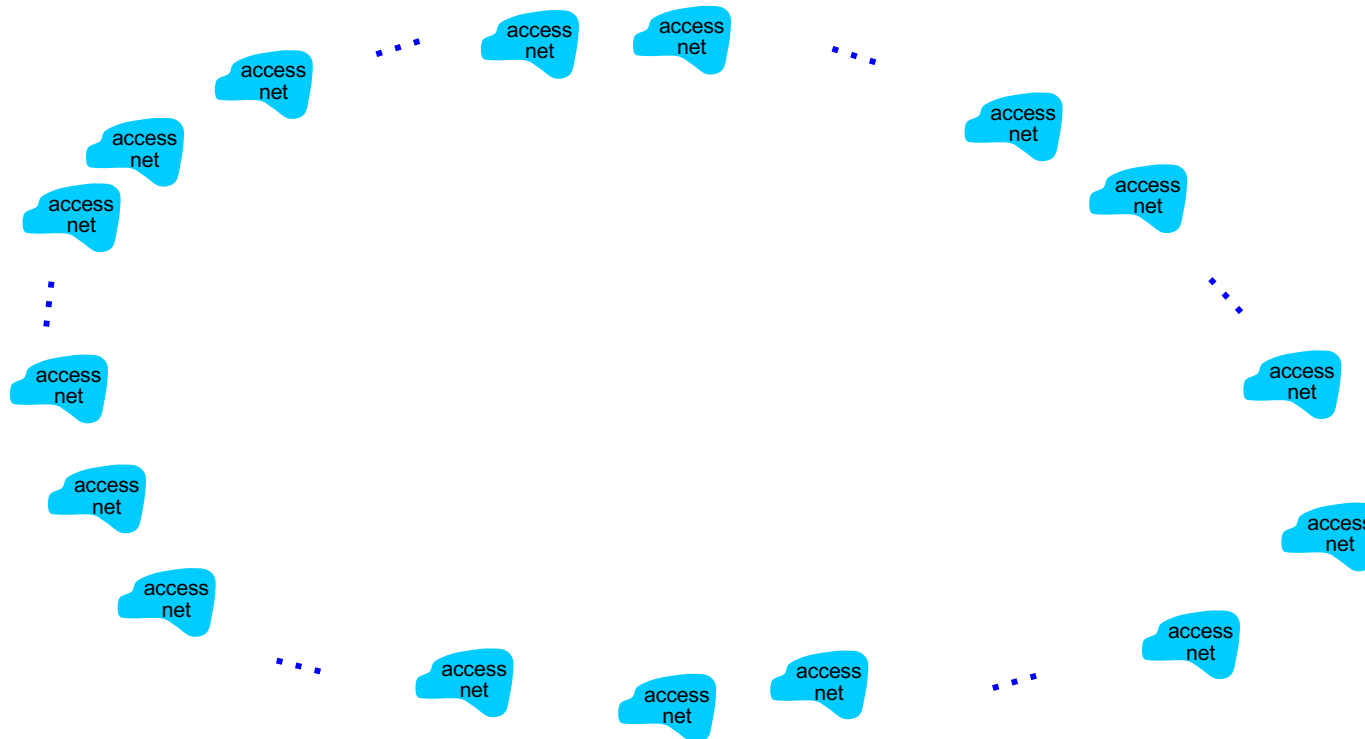


Internet Structure: A “network of networks”

- Hosts connect to Internet via access Internet Service Providers (ISPs)
 - residential, enterprise (company, university, commercial) ISPs
- Access ISPs in turn must be interconnected
 - so that any two hosts can send packets to each other
- Resulting network of networks is very complex
 - evolution was driven by economics and national policies
- Let's take a stepwise approach to describe current Internet structure

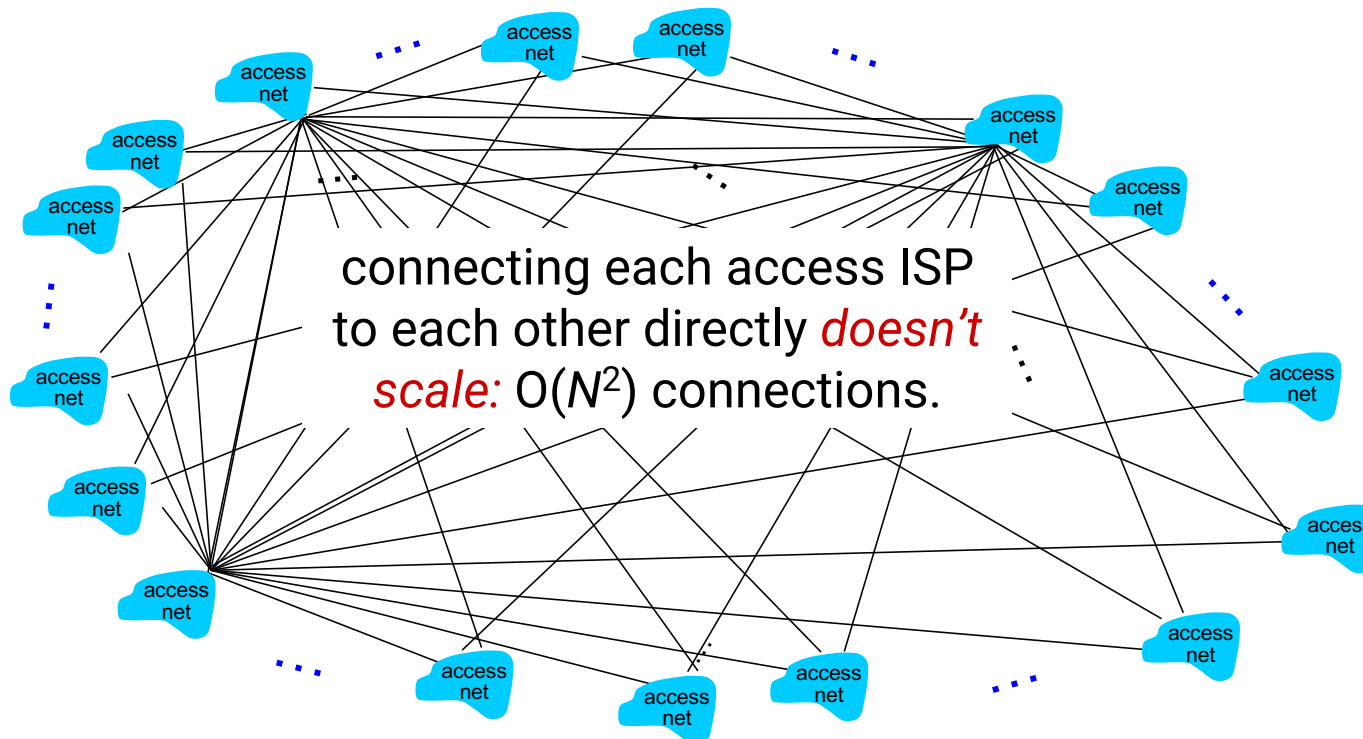
Internet structure: a “network of networks”

Question: given *millions* of access ISPs, how to connect them together?



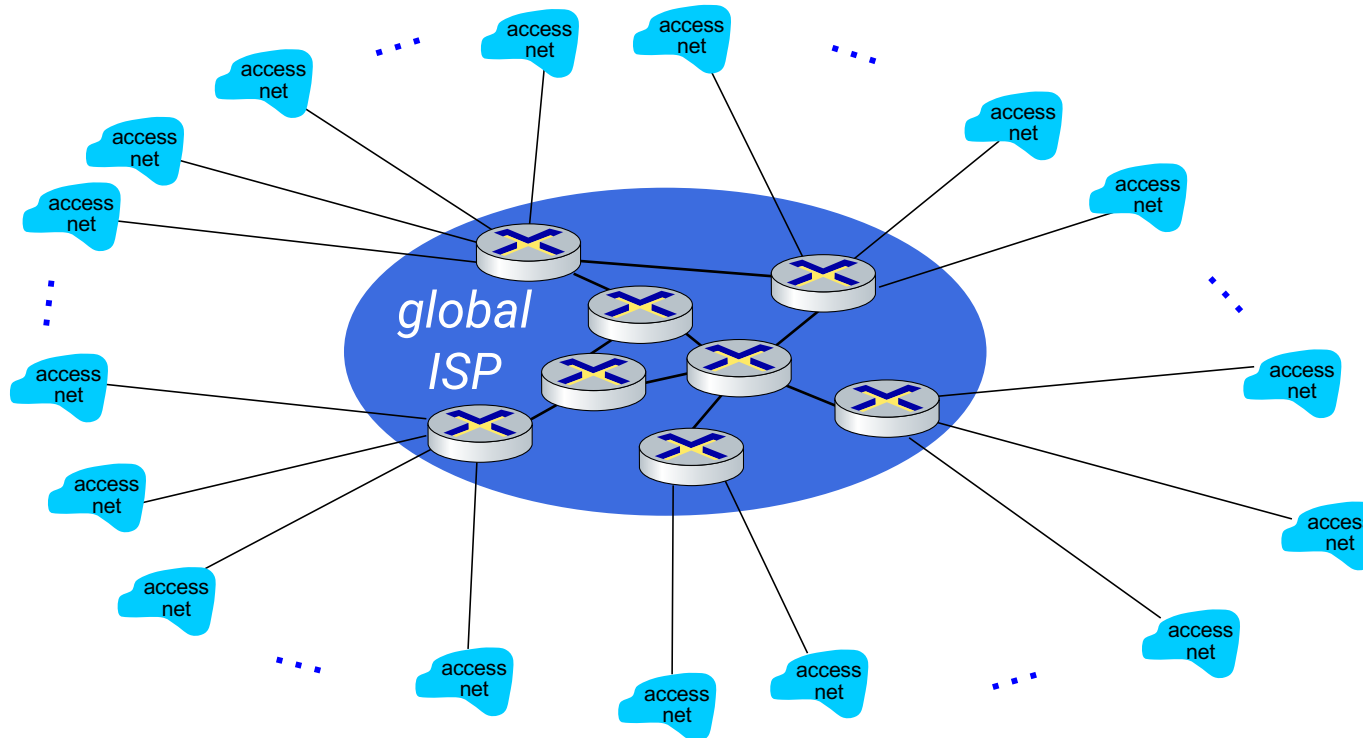
Internet structure: a “network of networks”

Question: given *millions* of access ISPs, how to connect them together?



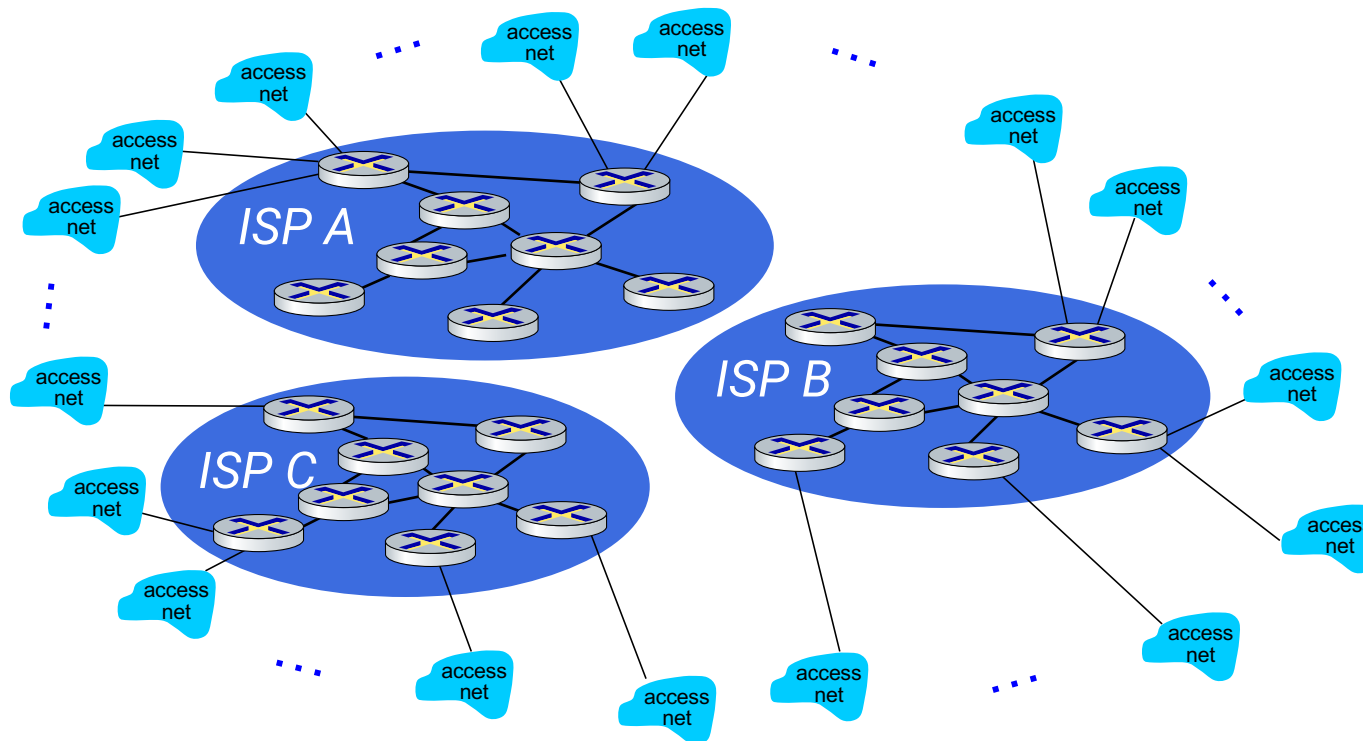
Internet structure: a “network of networks”

Option: connect each access ISP to one global transit ISP?
Customer and provider ISPs have economic agreement.



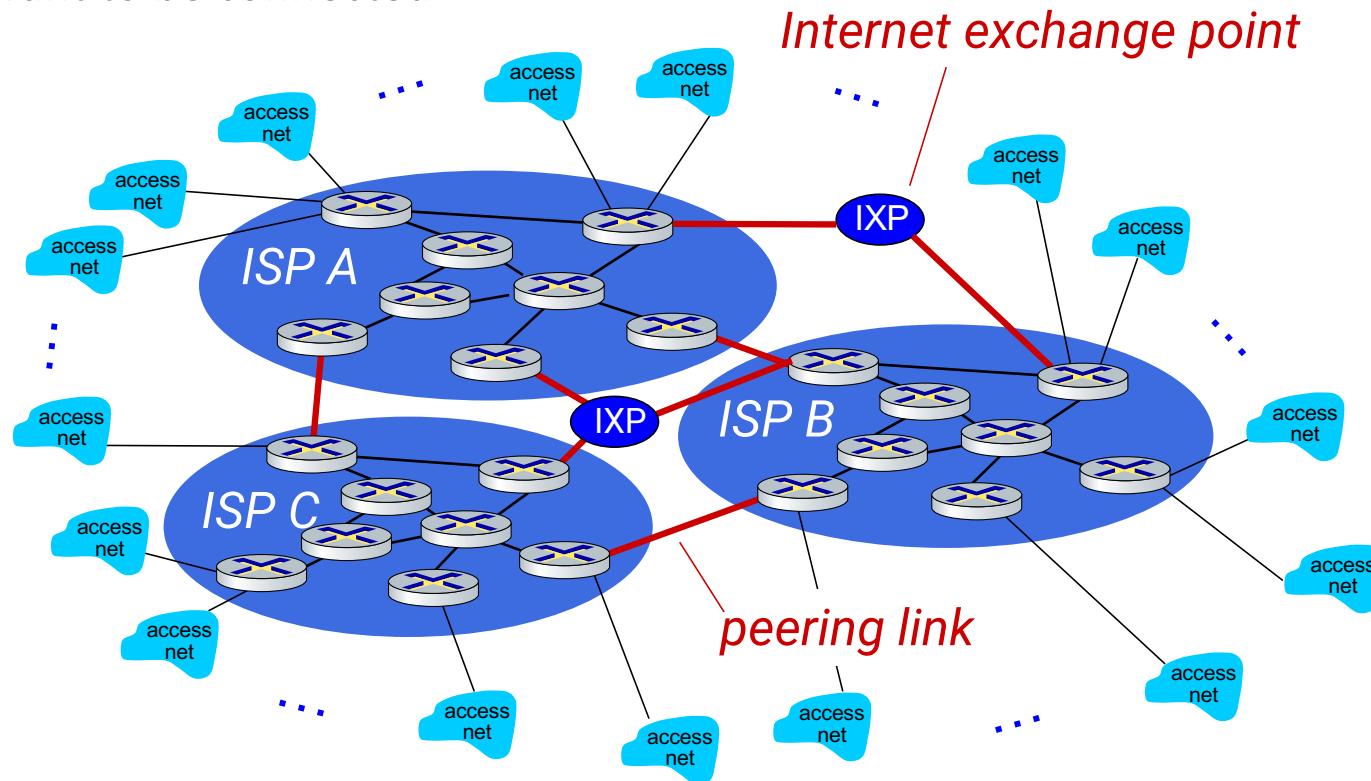
Internet structure: a “network of networks”

But if one global ISP is viable business, there will be competitors



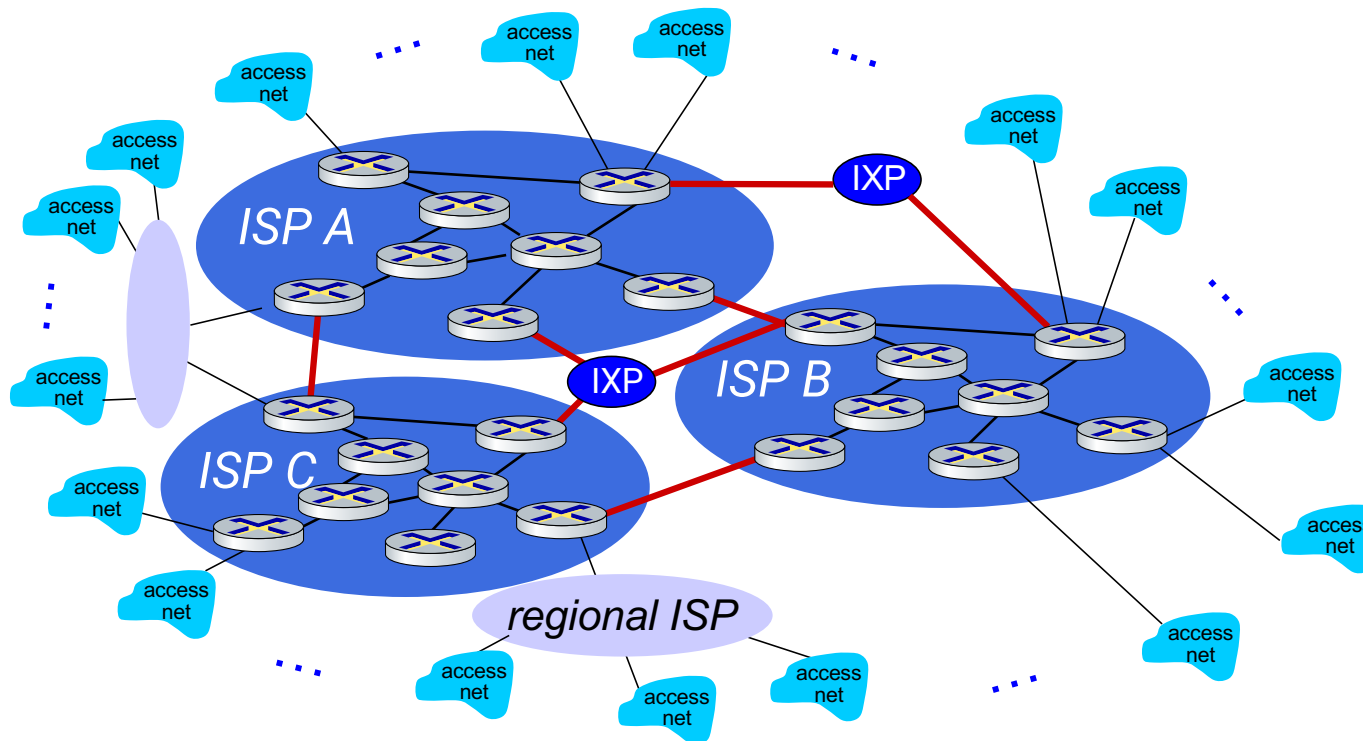
Internet structure: a “network of networks”

But if one global ISP is viable business, there will be competitors who will want to be connected



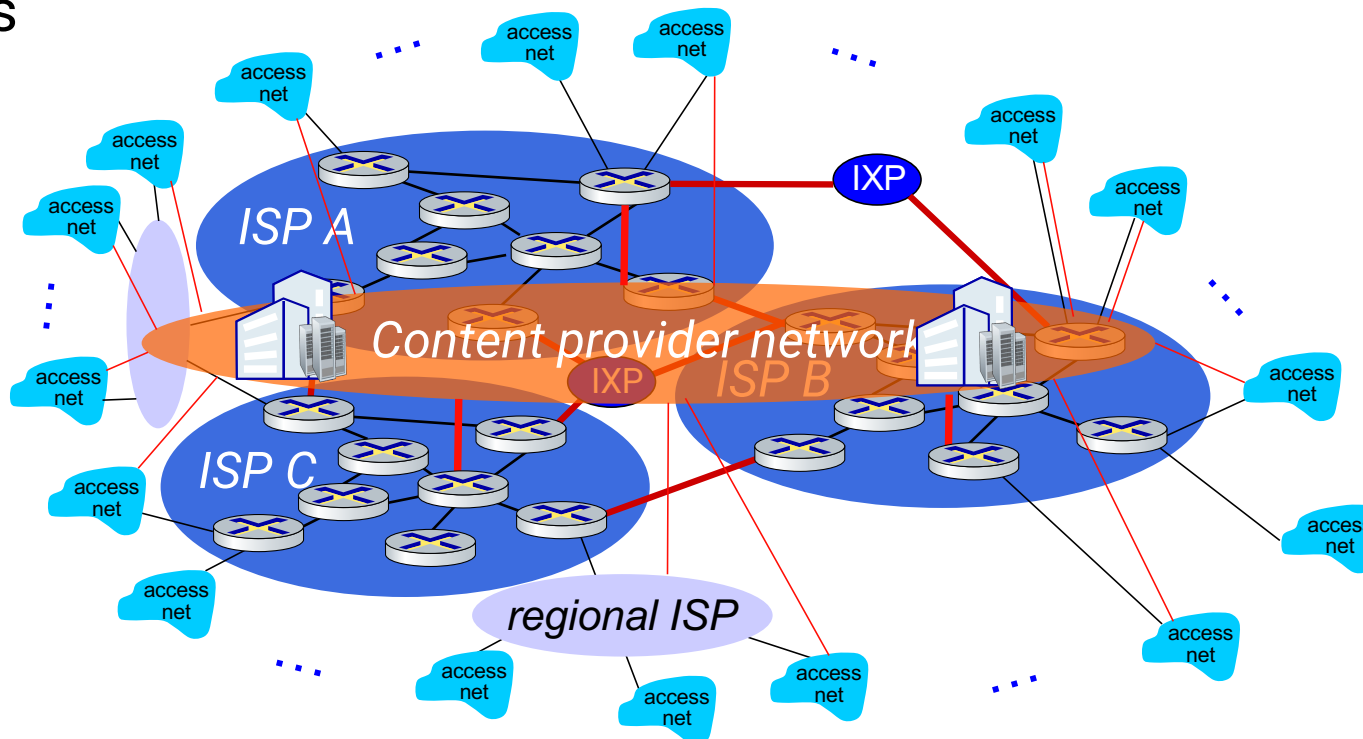
Internet structure: a “network of networks”

... and regional networks may arise to connect access nets to ISPs

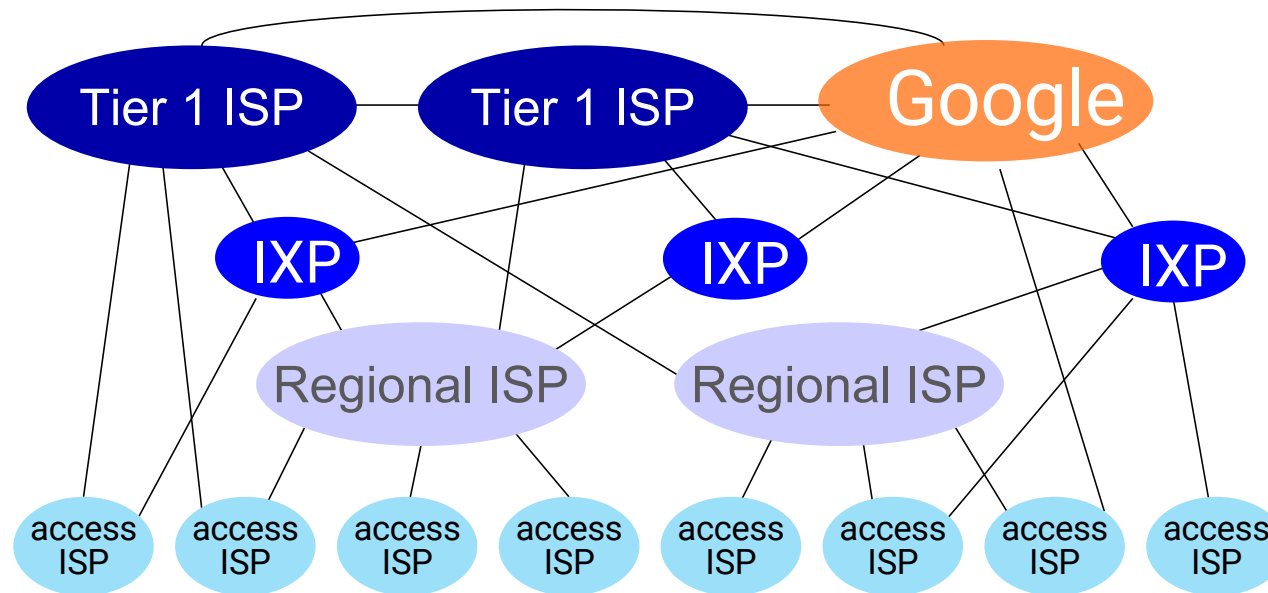


Internet structure: a “network of networks”

... and content provider networks (e.g., Google, Microsoft, Akamai) may run their own network, to bring services, content close to end users



Internet structure: a “network of networks”



At “center”: small # of well-connected large networks

- **“tier-1” commercial ISPs**: national & international coverage
- **content provider networks** (e.g., Google, Facebook): private network that connects its data centers to Internet, often bypassing tier-1, regional ISPs

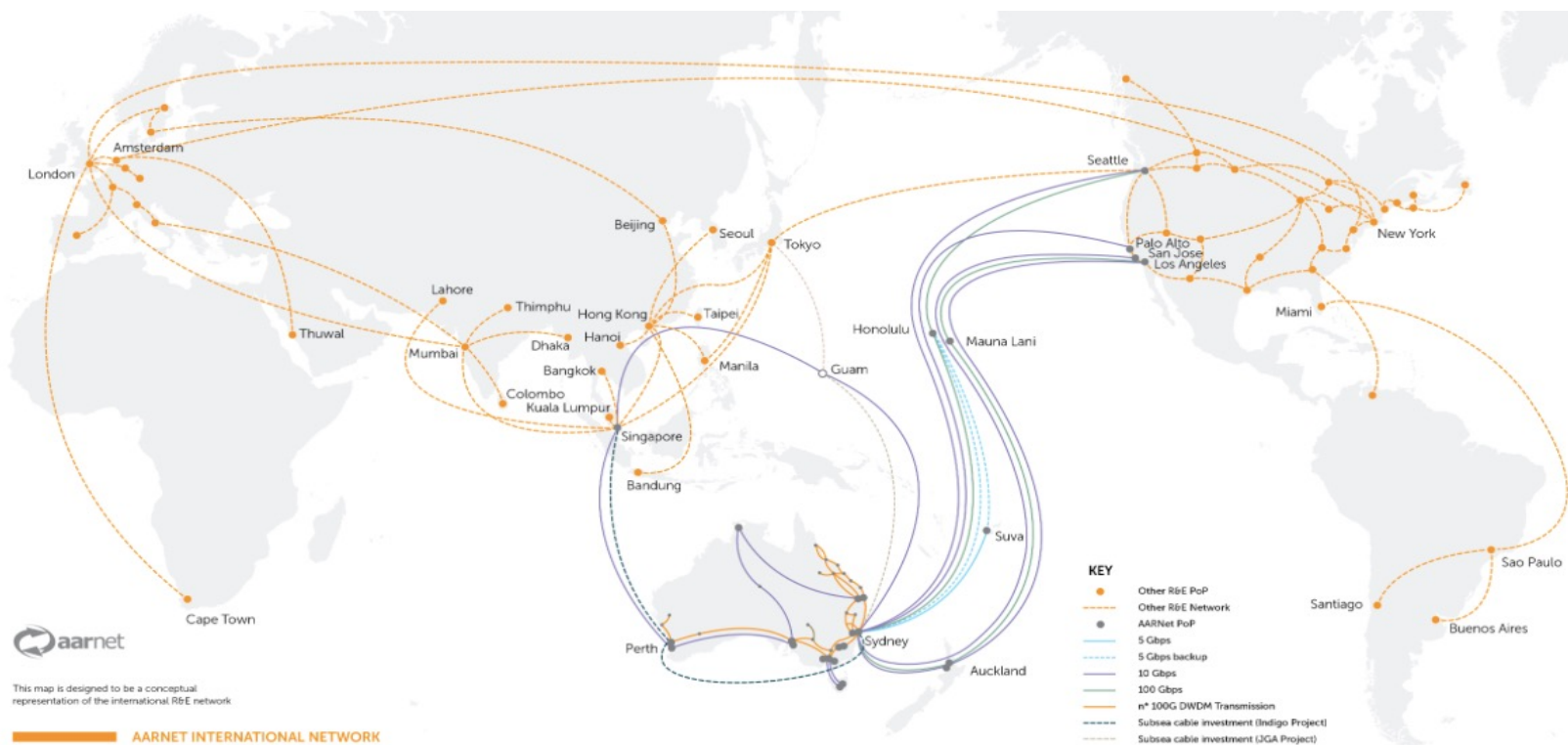
Tier-1 ISP Network map: Sprint (2019)



AARNET: Australia's Academic and Research Network

<https://www.aarnet.edu.au/>

<https://www.submarinecablemap.com>



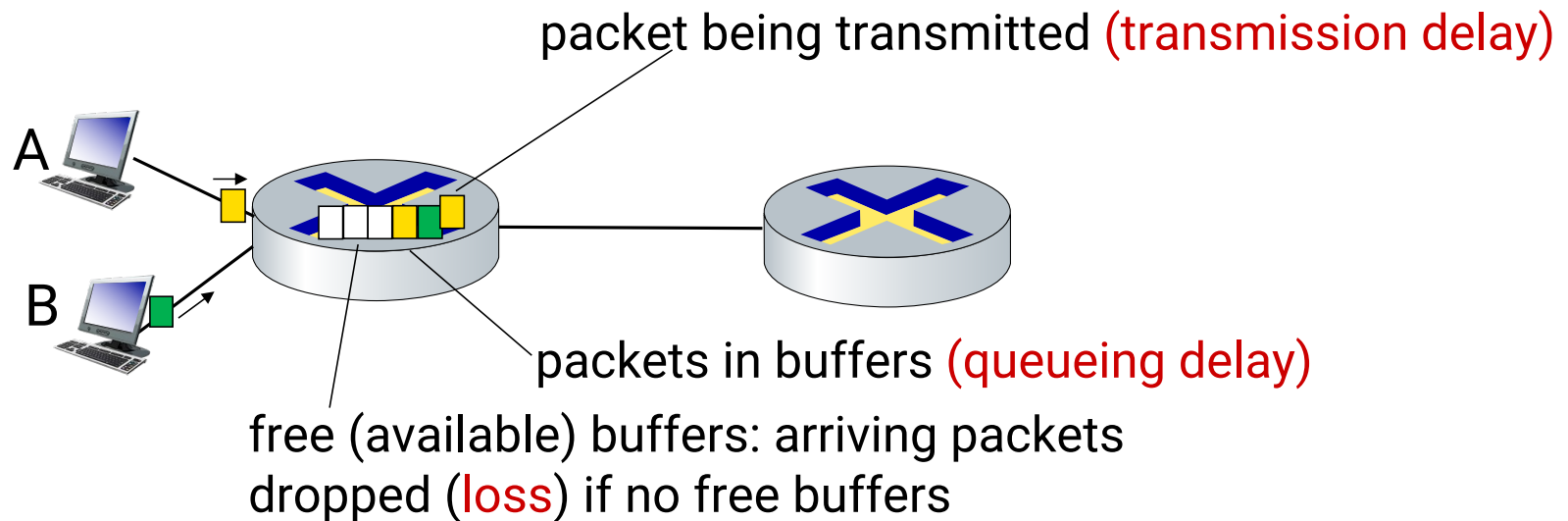
Introduction: Roadmap

- What is the Internet?
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- Performance: loss, delay, throughput
- Protocols layers, service models
- Security (self study, not on exam)
- History (self study, not on exam)

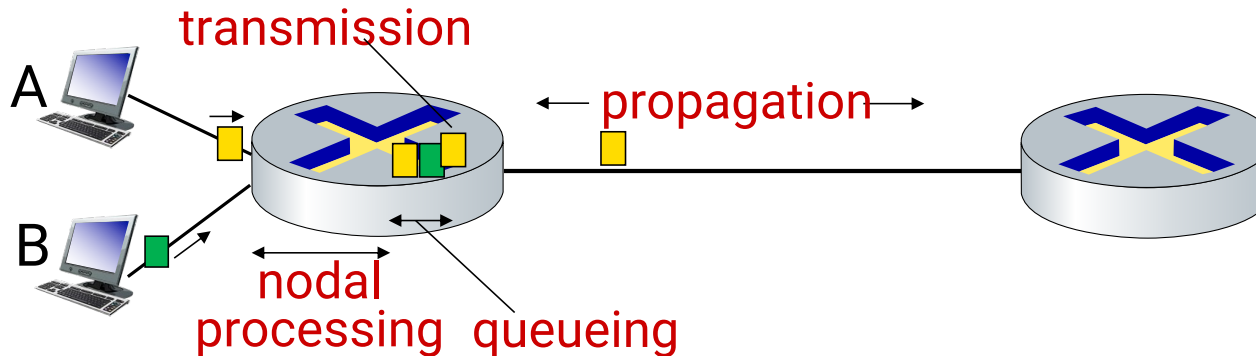
How do packet loss and delay occur?

Packets queue in router buffers

- packets queue, wait for turn
- arrival rate to link (temporarily) exceeds output link capacity: packet loss



Packet Delay: Four Sources



$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

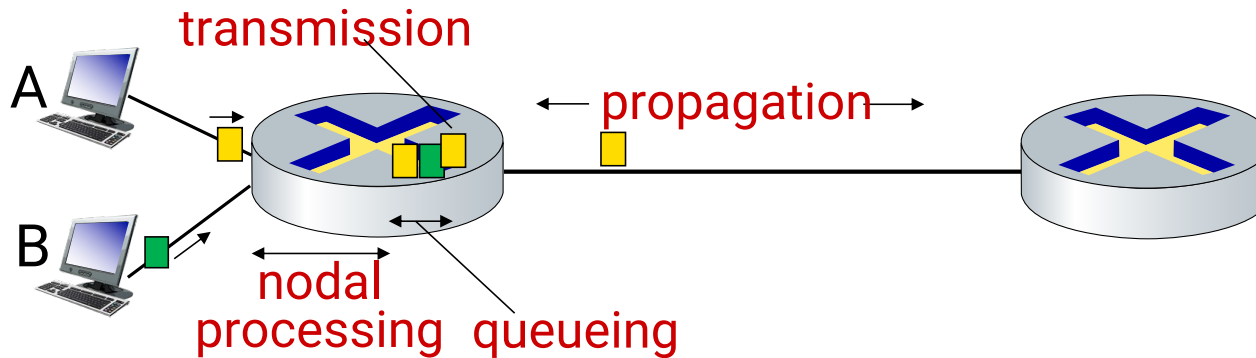
d_{proc} : nodal processing

- check bit errors
- determine output link
- typically < msec

d_{queue} : queueing delay

- time waiting at output link for transmission
- depends on congestion level of router

Packet Delay: Four Sources



$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

d_{trans} : transmission delay:

- L : packet length (bits)
- R : link transmission rate (bps)

▪ $d_{\text{trans}} = L/R$

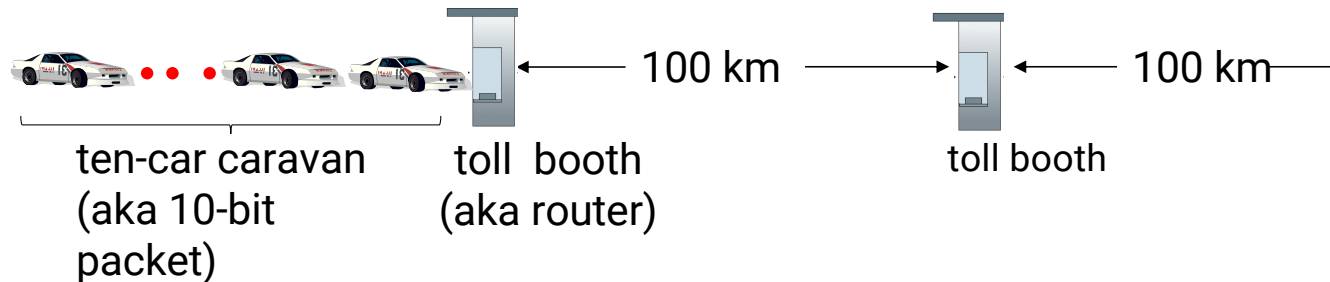
d_{prop} : propagation delay:

- d : length of physical link
- s : propagation speed ($\sim 2 \times 10^8$ m/sec)

▪ $d_{\text{prop}} = d/s$

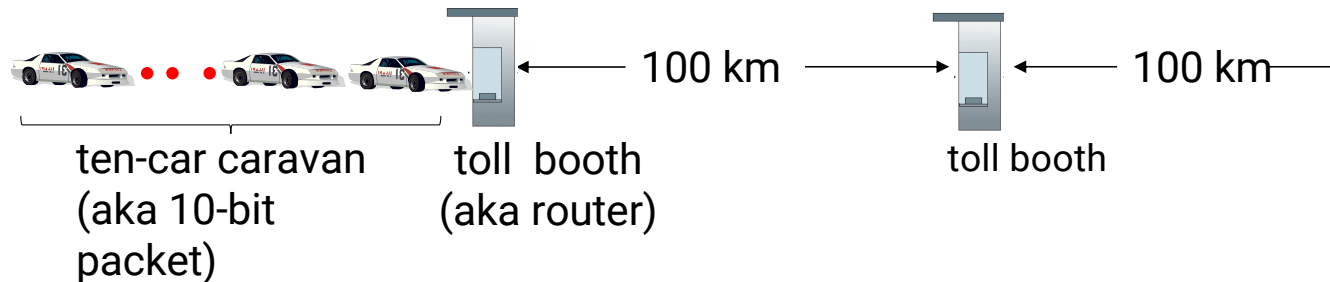
d_{trans} and d_{prop}
very different

Caravan Analogy



- cars “propagate” at 100 km/hr
- toll booth takes 12 sec to service car (bit transmission time)
- car ~ bit; caravan ~ packet
- **Q: How long until caravan is lined up before 2nd toll booth?**
- time to “push” entire caravan through toll booth onto highway
= $12 \times 10 = 120$ sec
- time for last car to propagate from 1st to 2nd toll booth:
 $100\text{km} / (100\text{km/hr}) = 1$ hr
- **A: 62 minutes**

Caravan Analogy

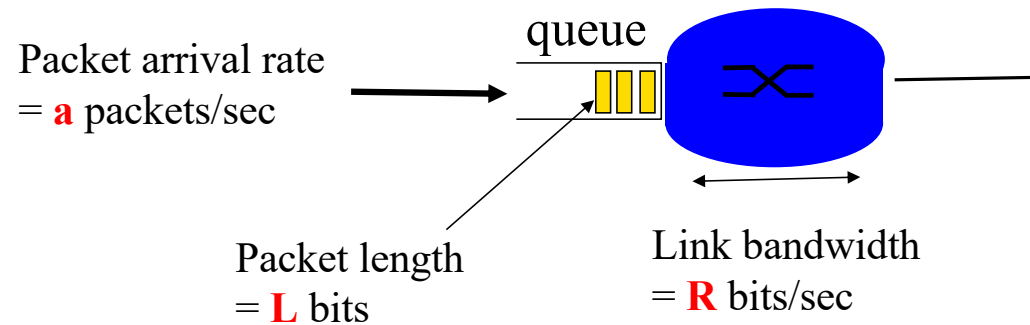


- suppose cars now “propagate” at 1000 km/hr
- and suppose toll booth now takes one min to service a car
- **Q: Will cars arrive to 2nd booth before all cars serviced at first booth?**

A: Yes! after 7 min, first car arrives at second booth; three cars still at first booth

Interactive Java Applet – Propagation vs transmission delay
<https://www2.tkn.tu-berlin.de/teaching/rn/animations/propagation/>

Queuing Delay: More insight



Every second: aL bits arrive to queue

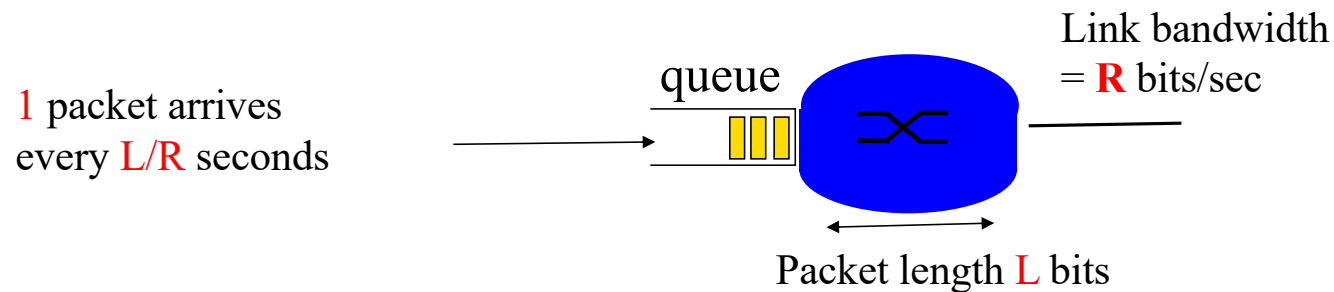
Every second: R bits leave the router

Question: what happens if $aL > R$?

Answer: queue will fill up, and packets will get dropped!!

aL/R is called **traffic intensity**

Queuing Delay: One Scenario



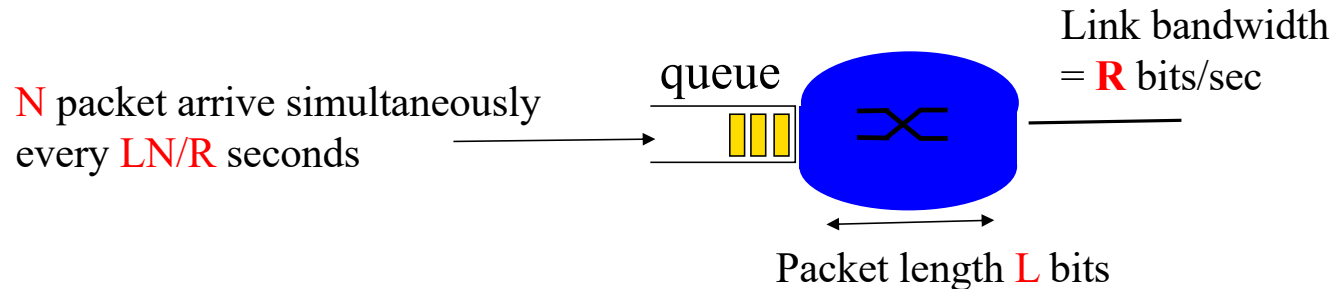
Arrival rate: $a = 1/(L/R) = R/L$ (packet/second)

Traffic intensity $= aL/R = (R/L) (L/R) = 1$

Average queueing delay $= 0$
(queue is initially empty)



Queueing Delay: Another Scenario



Arrival rate: $a = N/(LN/R) = R/L$ packet/second

Traffic intensity $= aL/R = (R/L) (L/R) = 1$

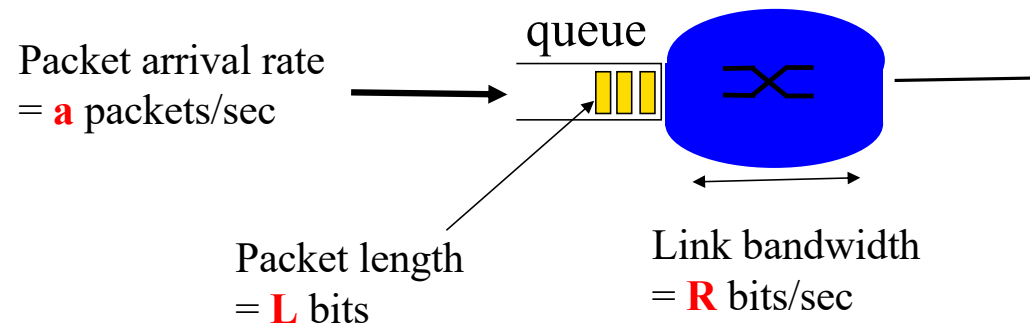
Average queueing delay (queue is empty at time 0) ?

$$\{0 + L/R + 2L/R + \dots + (N-1)L/R\}/N = L/(RN)\{1+2+\dots+(N-1)\} = L(N-1)/(2R)$$

Note: traffic intensity is same as previous scenario, but queueing delay is different

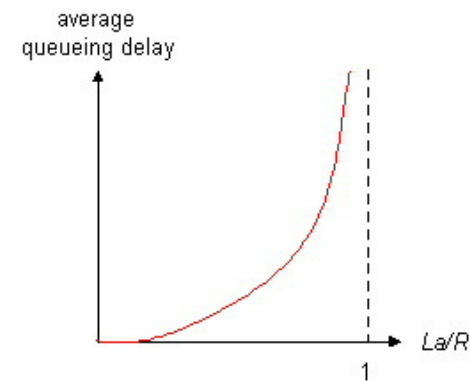


Queuing Delay: Typical behaviour

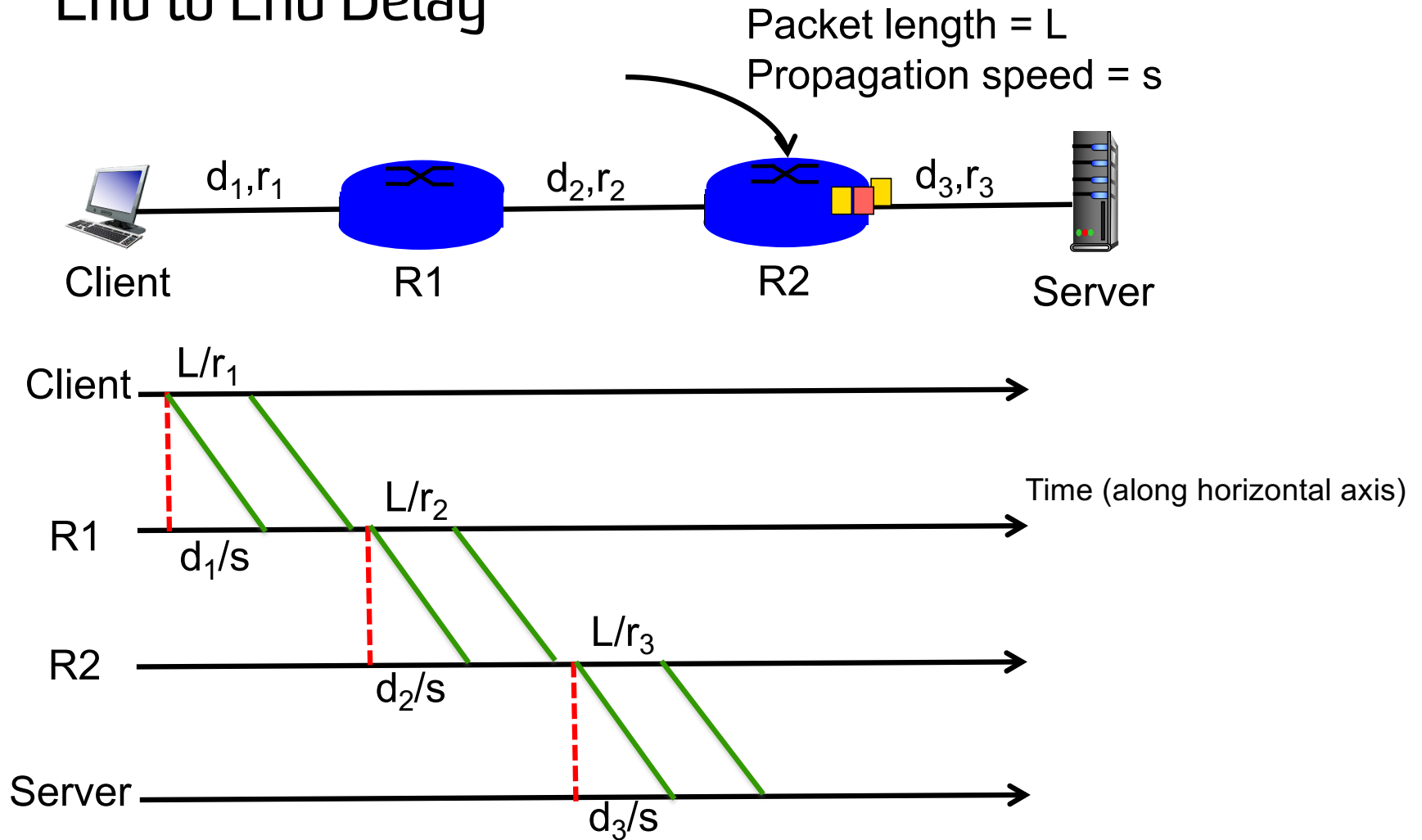


Interactive Java Applet:
<http://computerscience.unicam.it/marcantoni/reti/applet/QueuingAndLossInteractive/1.html>

- ❑ $\lambda a / R \sim 0$: avg. queueing delay small
- ❑ $\lambda a / R \rightarrow 1$: delays become large
- ❑ $\lambda a / R > 1$: more “work” than can be serviced, average delay infinite!
(this is when a is random!)

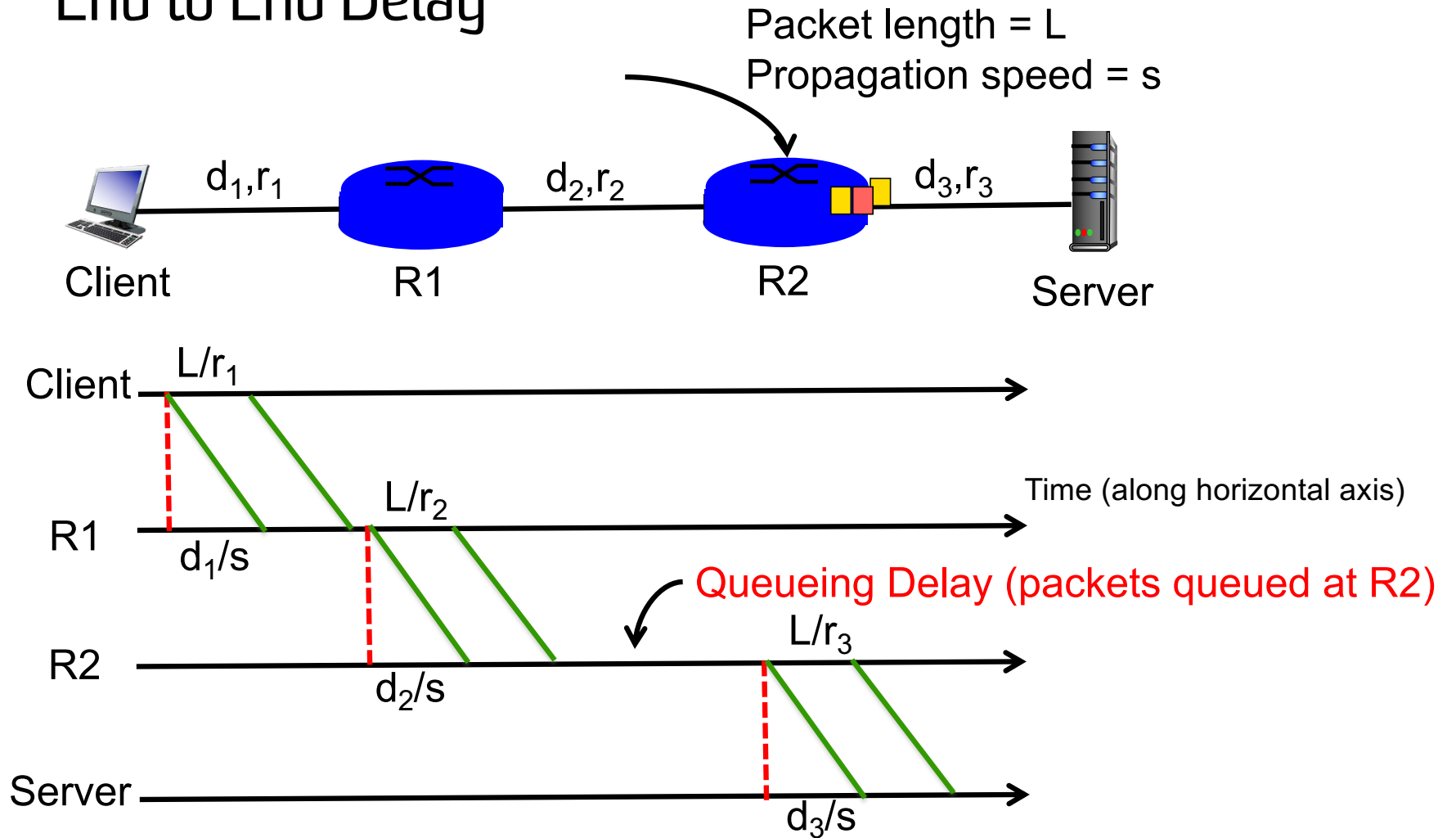


End to End Delay



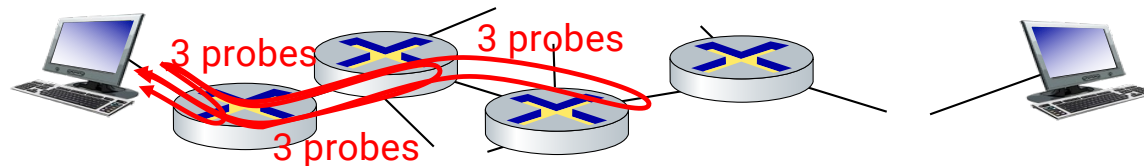
In the picture, $r_1 = r_2 = r_3$, you may wish to consider what happens when this is not the case

End to End Delay



“Real” Internet delays and routes

- what do “real” Internet delay & loss look like?
- **tracert** program: provides delay measurement from source to router along end-end Internet path towards destination. For all i :
 - sends three packets that will reach router i on path towards destination (with time-to-live field value of i)
 - router i will return packets to sender
 - sender measures time interval between transmission and reply



Real Internet delays and routes

traceroute: gaia.cs.umass.edu to www.eurecom.fr

3 delay measurements from
gaia.cs.umass.edu to cs-gw.cs.umass.edu

3 delay measurements
to border1-rt-fa5-1-0.gw.umass.edu

trans-oceanic link

looks like delays
decrease! Why?

* means no response (probe lost, router not replying)

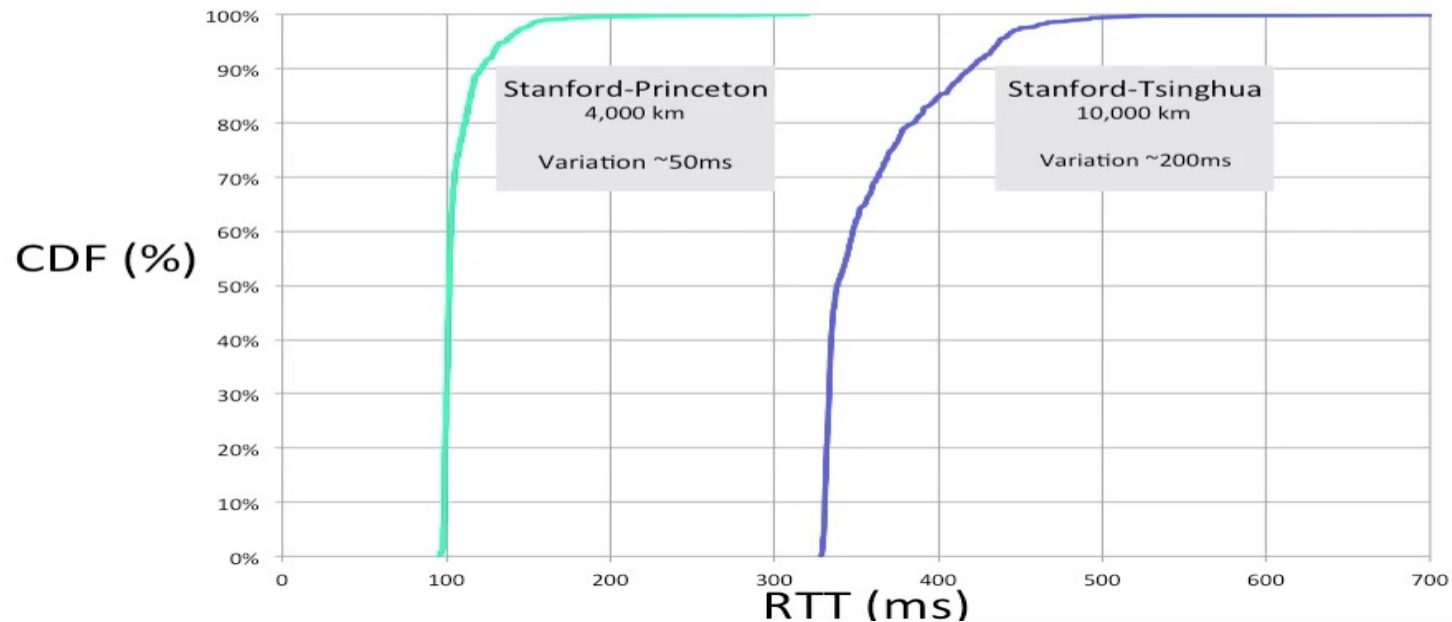
```
1 cs-gw (128.119.240.254) 1 ms 1 ms 2 ms
2 border1-rt-fa5-1-0.gw.umass.edu (128.119.3.145) 1 ms 1 ms 2 ms
3 cht-vbns.gw.umass.edu (128.119.3.130) 6 ms 5 ms 5 ms
4 jn1-at1-0-0-19.wor.vbns.net (204.147.132.129) 16 ms 11 ms 13 ms
5 jn1-so7-0-0-0.wae.vbns.net (204.147.136.136) 21 ms 18 ms 18 ms
6 abilene-vbns.abilene.ucaid.edu (198.32.11.9) 22 ms 18 ms 22 ms
7 nycm-wash.abilene.ucaid.edu (198.32.8.46) 22 ms 22 ms 22 ms
8 62.40.103.253 (62.40.103.253) 104 ms 109 ms 106 ms
9 de2-1.de1.de.geant.net (62.40.96.129) 109 ms 102 ms 104 ms
10 de.fr1.fr.geant.net (62.40.96.50) 113 ms 121 ms 114 ms
11 renater-gw.fr1.fr.geant.net (62.40.103.54) 112 ms 114 ms 112 ms
12 nio-n2.cssi.renater.fr (193.51.206.13) 111 ms 114 ms 116 ms
13 nice.cssi.renater.fr (195.220.98.102) 123 ms 125 ms 124 ms
14 r3t2-nice.cssi.renater.fr (195.220.98.110) 126 ms 126 ms 124 ms
15 eurecom-valbonne.r3t2.ft.net (193.48.50.54) 135 ms 128 ms 133 ms
16 194.214.211.25 (194.214.211.25) 126 ms 128 ms 126 ms
17 * * *
18 * * *
```

* Do some traceroutes from exotic countries at www.traceroute.org

“Real” delay variations

$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

End-to-end delay = sum of all d_{nodal} along the path



Quiz 1-6

Propagation delay depends on the size of the packet.

- A. True
- B. False



Quiz 1-7

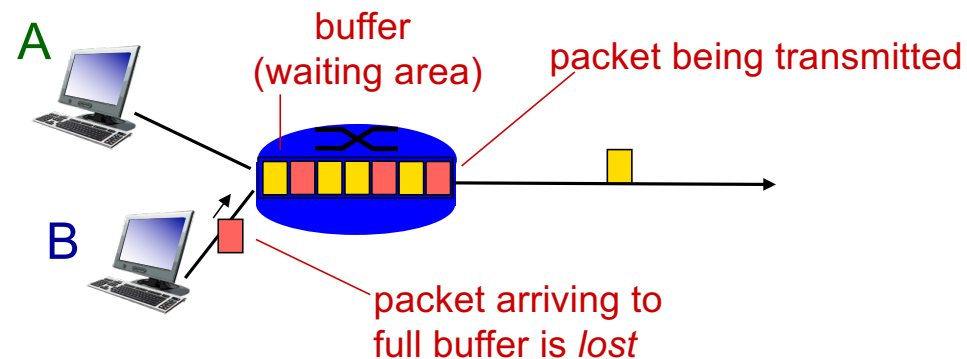
Consider a packet that has just arrived at a router. What is the correct order of the delays encountered by the packet until it reaches the next-hop router?

- A. Transmission, processing, propagation, queuing
- B. Propagation, processing, transmission, queuing
- C. Processing, queuing, transmission, propagation
- D. Queuing, processing, propagation, transmission



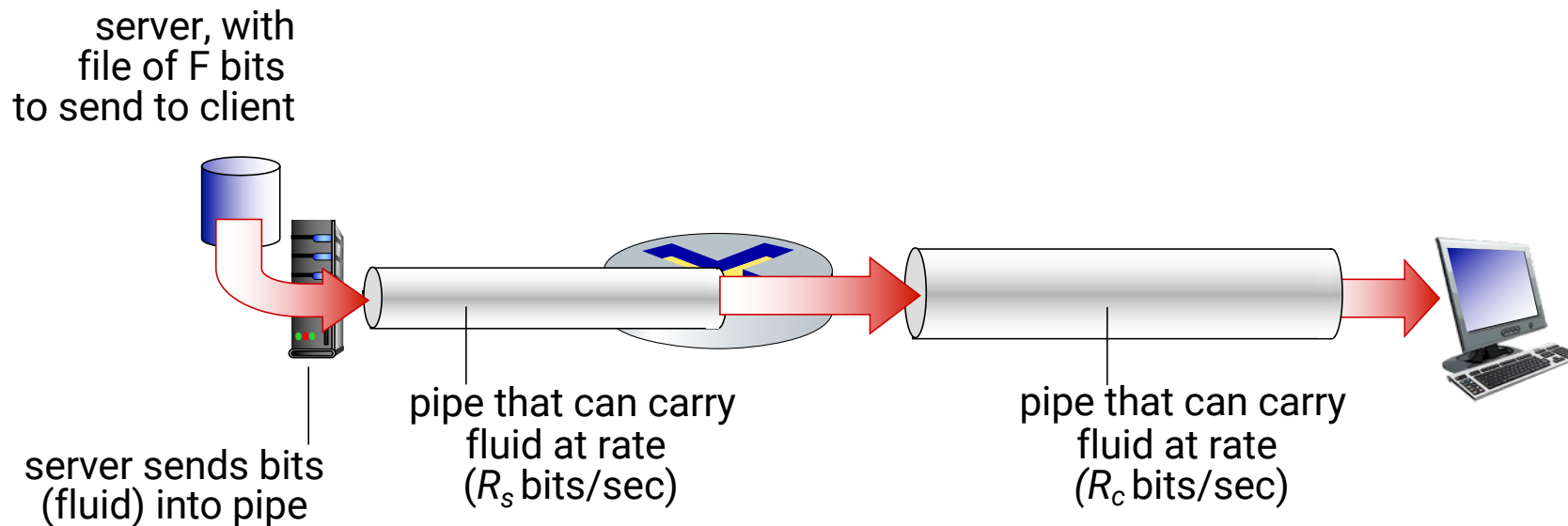
Packet loss

- Queue (aka buffer) preceding link in buffer has finite capacity
- Packet arriving to full queue dropped (aka lost)
- Lost packet may be retransmitted by previous node, source end system, or not at all



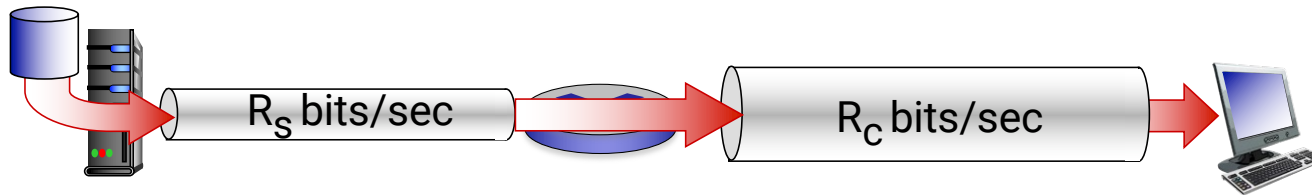
Throughput

- Throughput: rate (bits/time unit) at which bits are being sent from sender to receiver
 - **instantaneous**: rate at given point in time
 - **average**: rate over longer period of time

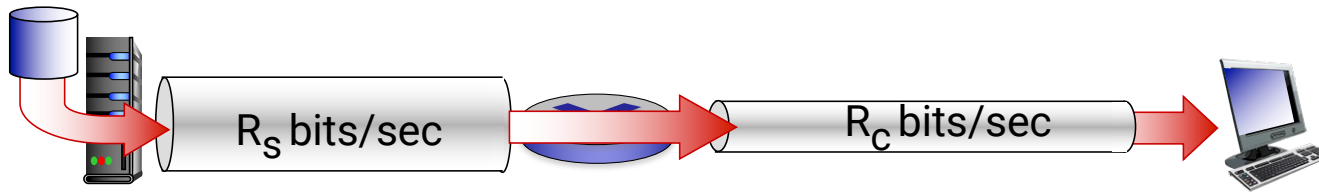


Throughput

$R_s < R_c$ What is average end-end throughput?



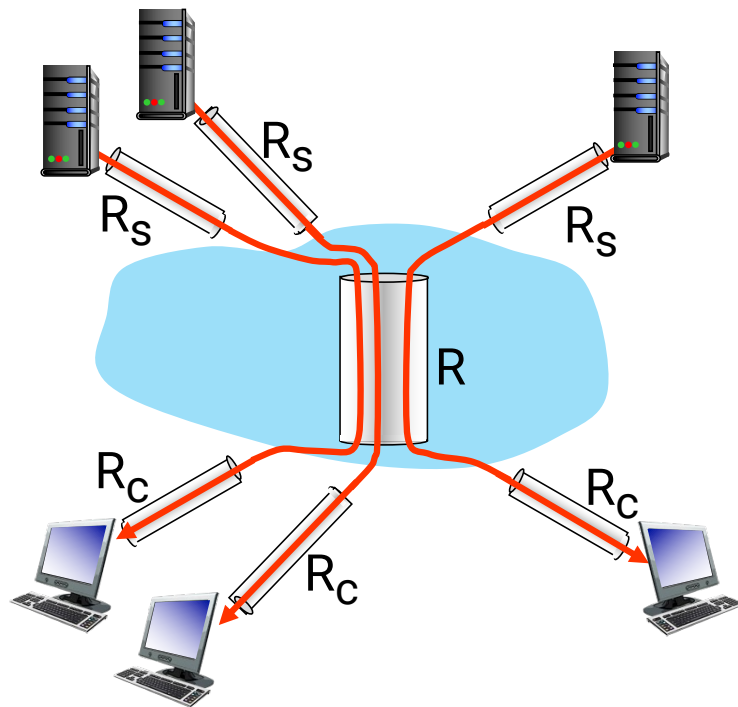
$R_s > R_c$ What is average end-end throughput?



bottleneck link

link on end-end path that constrains end-end throughput

Throughput: network scenario



10 connections (fairly) share
backbone bottleneck link R bits/sec

- per-connection end-end throughput: $\min(R_c, R_s, R/10)$
- in practice: R_c or R_s is often bottleneck

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- History (self study, not on exam)

Binomial Probability Distribution (self-study)

A fixed number of observations (trials), n

- E.g., 5 tosses of a coin

Binary random variable

- E.g., head or tail in a coin toss
- Often called as success or failure
- Probability of success is p and failure is $(1-p)$

Constant probability for each observation

Binomial Distribution: Example

Q: What is the probability of observing exactly 3 heads in a sequence of 5 coin tosses

A:

- One way to get exactly 3 heads is: HHHTT
- Probability of this sequence occurring = $(1/2) \times (1/2) \times (1/2) \times (1-1/2) \times (1-1/2) = (1/2)^5$
- Another way to get exactly 3 heads is: THHHT
- Probability of this sequence occurring = $(1-1/2) \times (1/2) \times (1/2) \times (1/2) \times (1-1/2) = (1/2)^5$
- How many such unique combinations exist?

Binomial Distribution: Example

$\binom{5}{3}$

ways to
arrange 3
heads in
5 trials

Outcome	Probability
THHHT	$(1/2)^3 \times (1/2)^2$
HHHTT	$(1/2)^3 \times (1/2)^2$
TTHHH	$(1/2)^3 \times (1/2)^2$
HTTHH	$(1/2)^3 \times (1/2)^2$
HHTTH	$(1/2)^3 \times (1/2)^2$
THTHH	$(1/2)^3 \times (1/2)^2$
HTHTH	$(1/2)^3 \times (1/2)^2$
HHTHT	$(1/2)^3 \times (1/2)^2$
THHTH	$(1/2)^3 \times (1/2)^2$
<u>HTHHT</u>	<u>$(1/2)^3 \times (1/2)^2$</u>
10 arrangements $\times (1/2)^3 \times (1/2)^2$	

The probability
of each unique
outcome (note:
they are all
equal)

${}^5C_3 = 5!/3!2! = 10$

$$P(3 \text{ heads and } 2 \text{ tails}) = 10 \times (1/2)^5 = 0.3125$$

Binomial Distribution

Note the general pattern emerging → if you have only two possible outcomes (call them 1/0 or yes/no or success/failure) in n independent trials, then the probability of exactly X “successes”=

The diagram shows the binomial distribution formula $\binom{n}{X} p^X (1-p)^{n-X}$ enclosed in a purple rectangular box. Four arrows point from descriptive text to parts of the formula: one from ' n ' to ' $n = \text{number of trials}$ ', one from ' X ' to ' $X = \# \text{ successes out of } n \text{ trials}$ ', one from ' p ' to ' $p = \text{probability of success}$ ', and one from ' $1-p$ ' to ' $1-p = \text{probability of failure}$ '.

$$\binom{n}{X} p^X (1-p)^{n-X}$$

$n = \text{number of trials}$

$X = \#$
successes
out of n
trials

$p =$
probability of
success

$1-p = \text{probability}$
of failure

Packet switching versus circuit switching

Let's revisit the earlier problem

N = 35 users

$$\begin{aligned}\text{Prob}(\# \text{ active users} > 10) &= 1 - \text{Prob}(\# \text{ active} = 10) \\ &\quad - \text{Prob}(\# \text{ active} = 9) \\ &\quad - \text{Prob}(\# \text{ active} = 8) \\ &\quad \dots \\ &\quad - \text{Prob}(\# \text{ active} = 0)\end{aligned}$$

$$\text{where } \text{Prob}(\# \text{ active} = 10) = C(35, 10) \times 0.1^{10} \times 0.9^{25}$$

$$\text{Prob}(\# \text{ active users} > 10) = 0.0004 \text{ (approx)}$$